

Additionality and Asymmetric Information in Environmental Markets: Evidence from Conservation Auctions[†]

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Mechanisms that aim to reduce environmental degradation at low cost can be undermined when participants' conservation actions are not marginal to the incentive—or “additional”—as the lowest-cost participants may not be the highest social value. We investigate this challenge in the Conservation Reserve Program's auction mechanism for ecosystem services, linking bids to satellite-derived land use. Three-quarters of marginal auction winners are not additional. The heterogeneity in counterfactual land use introduces adverse selection. We develop a model of bidding and additionality to quantify welfare implications. Alternative auctions increase efficiency by using scoring rules that incorporate expected land use impacts. (JEL D44, D82, Q24, Q57, R14)

Land use change contributes 11 percent of global CO₂ emissions (Friedlingstein et al. 2026) and leads to biodiversity loss, water pollution, and erosion (Dirzo et al. 2014; Vörösmarty et al. 2010; Borrelli et al. 2017). Market mechanisms for the provision of environmental services aim to combat this degradation at low cost, but many believe that existing mechanisms with voluntary participation have failed to achieve this objective (Anderson 2012; West, Börner, Sills, and Kontoleon 2020; Jones and Lewis 2023). A leading explanation is the possibility that participants may have engaged in the incentivized action absent an incentive. The notion of “additionality,” defined as the likelihood that an action is marginal to an incentive, is therefore a central challenge to the design and success of many environmental markets (Engel, Pagiola, and Wunder 2008; West et al. 2023).

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When does the challenge of additionality undermine the performance of mechanisms incentivizing conservation, and how can market rules be designed to achieve environmental objectives at low cost? We explore these questions by analyzing the challenge of additionality as a problem of asymmetric information. Social welfare in markets for environmental conservation depends on both a landowner's unobserved additionality and her private cost of complying with the market requirements. Market mechanisms, however, screen only on the latter and therefore may not necessarily implement an efficient allocation (Akerlof 1970). In this paper, we use this perspective to analyze, test, and quantify this potential inefficiency and to examine remedies in alternative market designs.

We study the United States Department of Agriculture's (USDA) Conservation Reserve Program (CRP), one of the oldest and largest payments for ecosystem services (PES) mechanisms in the world. The CRP incentivizes agricultural land retirement and conservation actions via procurement auctions of conservation contracts. CRP contracts pay landowners \$1.6–\$1.8 billion per year to take cropland out of production and to plant grass mixes, plant or maintain trees, or establish habitats for a duration of ten years. Combining administrative data and satellite-derived land use classifications, we construct a dataset that links landowners' multidimensional bids in the CRP-scoring auction to their land use, which we use to measure additionality.

We first define additionality and analyze market efficiency in a stylized framework that builds on Einav, Finkelstein, and Cullen (2010). Landowners differ in two dimensions. The first is their cost of accepting a CRP contract, which includes the forgone option value of cropping over the contract period and the hassle costs of complying with program requirements. The second is their conservation behavior without the contract, which determines their additionality. The social value of contracting depends on a landowner's cost and additionality, but her choices depend only on her cost and the market incentive. A landowner's cost of accepting the contract may be positively correlated with her additionality, as landowners who are likely to conserve regardless of the CRP have lower opportunity costs of contracting, introducing adverse selection in the market. These differences between socially optimal and landowner-optimal choices could limit the implementability of efficient allocations and undermine the performance of standard procurement mechanisms (Manelli and Vincent 1995). These challenges may be remedied by alternative market designs that narrow this gap. Ultimately, the extent of additionality and adverse selection and together their quantitative implications for the performance and design of markets for environmental services are empirical questions.

We begin by examining the extent of additionality in our setting. We use the discontinuity in contracting around the winning score in the procurement auction to evaluate additionality at the margin of contract awards. We find that, as incentivized by the CRP, landowners substitute from agriculture to natural vegetation and grasslands upon contracting. However, only about one-quarter of marginal winners are additional, which we calculate by comparing the regression discontinuity treatment effect to the magnitude of land contracting at the margin. In other words, three-quarters of landowners at the margin of contract awards would have conserved without a CRP contract. However, the status quo auction implicitly assumes all landowners are additional in the design of its scoring rule (Claassen, Horowitz, Duquette, and Ueda 2014).

To test for adverse selection, we correlate heterogeneity in additionality with heterogeneity in costs reflected in landowners' bids. We make two assumptions—perfect compliance and no spillovers, both of which we test and validate—to obtain a landowner-specific measure of additionality for all rejected bidders (82 percent in the most restrictive auction). We examine the correlation between additionality and bids following tests for asymmetric information in insurance markets (Chiappori and Salanié 2000) and auctions (Hendricks and Porter 1988). We document substantial heterogeneity in additionality and a positive correlation between additionality and bids, which persist even conditional on a rich set of observed characteristics, indicating the presence of adverse selection in the market. We also highlight that heterogeneity in additionality is predicted by landowners' choice of conservation contract in the mechanism and by the observed characteristic of soil productivity.

These facts demonstrate that additionality and asymmetric information are relevant; to quantify welfare implications and evaluate the performance of counterfactual market designs, we develop and estimate a joint model of bidding and additionality. In the CRP auction, landowners submit multidimensional bids on a menu of heterogeneous contracts, which are ranked by a scoring rule. First, we infer the costs of contracts from revealed preferences in optimal bidding. Second, we model additionality with a conditional expectation function that relates land use to both observed characteristics and unobserved landowner costs. This conditional expectation function is the component of the model that captures the possibility of inefficient or adverse selection.

We estimate the model in three steps. The first two steps adapt standard procedures for the empirical analysis of auctions (Guerre, Perrigne, and Vuong 2000; Hortaçsu 2000; Hortaçsu and McAdams 2010; Agarwal, Li, and Somaini 2023). First, we estimate bidder beliefs via simulation. Second, we estimate bidder costs via revealed preferences in optimal bidding. In the final step, we estimate additionality, including how it varies with unobserved landowner costs, by matching observed land use among rejected bidders and the correlation between land use and landowner characteristics and optimal bids. We use our estimates of additionality to calculate the social benefits of contracting based on valuations of environmental services from the CRP literature and the USDA's revealed preferences across landowners and contracts implied by the scoring rule.

Using these estimates, we first examine whether the existence of a market for conservation increases social welfare. The answer to this question is theoretically ambiguous. We investigate it empirically in the context of a hypothetical uniform market for the base conservation contract. We find that, despite landowners who are not additional and the presence of adverse selection, there exist social welfare gains of \$14.37 per acre-year under the socially optimal uniform price. Second, we evaluate the status quo auction mechanism. We estimate social welfare gains of \$120 million per auction, relative to no market. However, while positive, the status quo auction implements only a fraction (12 percent) of the social welfare gains of an efficient allocation. This allocation determines contract awards based on both landowners' costs and expected social benefits, which depend on additionality.

Implementing the efficient allocation with an incentive-compatible auction may not be possible (Myerson 1981). Because they are less additional, lower-cost landowners are not always higher social value, and our estimates imply that the allocation

rule for this efficient benchmark need not be monotone in bidder cost. We instead propose and evaluate alternative auctions with scoring rules that trade off bidders' costs against both their conservation actions' heterogeneous social benefits and their expected additionality. This differs from the status quo scoring rule, which does not consider the latter. Alternative scoring rules adjust asymmetry across bidder observables and scores across the menu of contracts based on predictions of additionality. Our approach redesigns the auction to cost-effectively impact conservation, given that market participants are with some probability not additional.

Simple modifications to the auction's scoring rule close the gap between the status quo and efficient allocation by 37 percent, increasing social welfare by over \$300 million per auction. Almost all gains are due to changes that incorporate landowner additionality through modifications to the scoring rule. A large share of these gains accrue through simple changes, including a uniform adjustment to the scoring rule and its use to reduce the market's size. Further gains accrue by using the rule to differentiate among heterogeneously additional landowners. These gains could be implemented in practice by the USDA. More broadly, we highlight that successful market design considers both the heterogeneity in private costs that determine choices and the implications of these choices for land use impacts and social welfare.

Related Literature.—Our primary contributions are to develop a framework to evaluate the implications of additionality for social welfare under current and counterfactual market designs and to provide credible estimates of the extent of additionality and evidence of adverse selection in a large-scale, mature market for conservation. Our regression discontinuity estimates of additionality contribute to a literature estimating treatment effects of PES (Jack 2013; Alix-Garcia, Sims, and Yañez-Pagans 2015; Jayachandran et al. 2017; West, Börner, Sills, and Kontoleon 2020; Rosenberg and Pratt 2024) and offset markets (Calel, Colmer, Dechezleprêtre, and Glachant 2025). Our framework builds on theoretical analyses of asymmetric information (van Benthem and Kerr 2013; Mason and Plantinga 2013; Li, Ashlagi, and Lo 2022) and empirical tests for selection (Montero 1999; Jack 2013; Jack and Jayachandran 2018; Chen, Ryan, and Xu 2025) in environmental incentive programs and offset markets. Our research also relates broadly to the design of environmental incentive programs and other voluntary regulation (Einav, Finkelstein, Ji, and Mahoney 2022; Ito, Ida, and Tanaka 2023) and complements work studying other sources of inefficiency in markets for environmental conservation (Harstad 2016; Aronoff and Rafey 2026).

We contribute to a literature studying quality concerns in procurement auctions (Manelli and Vincent 1995; Decarolis 2014; Lopomo, Persico, and Villa 2023; Carril, Gonzalez-Lira, and Walker 2026), where we provide an empirical framework to evaluate alternative auction designs in the presence of adverse selection on bidder quality (additionality in our setting). This empirical framework draws on a large literature studying selection in insurance markets (Akerlof 1970; Chiappori and Salanié 2000; Einav, Finkelstein, and Cullen 2010; Bundorf, Levin, and Mahoney 2012; Marone and Sabety 2022). Our model and estimation strategy also use techniques from a literature advancing the empirical analysis of auctions (Guerre, Perrigne, and Vuong 2000; Hortaçsu 2000; Hortaçsu and McAdams 2010; Agarwal, Li, and Somaini 2023). We draw on and extend existing work on scoring and other multidimensional auctions (Che 1993; Asker and Cantillon 2008; Lewis and Bajari

2011; Kong, Perrigne, and Vuong 2022; Allen, Clark, Hickman, and Richert 2023; Bolotnyy and Vasserman 2023) to incorporate discrete bidding, a nonlinear scoring rule, and a correlation between additionality and bidder costs.

I. Theoretical Framework

A. Model

There exists a continuum of landowners, indexed by i , each making a decision to contract, $x_i \in \{0, 1\}$, to obtain a transfer, p . In Section IV, we adapt this framework to a finite number of landowners bidding for contracts in a procurement auction with a quantity constraint.

The contract involves a promise to provide an environmental service ($a_i = 1$) versus not ($a_i = 0$). In our setting, $a_i = 1$ denotes agricultural land retirement (conservation) and $a_i = 0$ denotes cropping. The action $a_i = 1$ generates social benefits from positive environmental externalities, $B > 0$.

Define a_{i1} as landowner i 's action when $x_i = 1$ and a_{i0} as her action when $x_i = 0$. We assume perfect compliance, so $a_{i1} = 1$. Because B is generated whenever i chooses $a_i = 1$, regardless of contract choice x_i , the benefit of contracting with i is only the *incremental value* $B \cdot (1 - a_{i0})$. a_{i0} is unobserved whenever $x_i = 1$ and is therefore noncontractible.

A landowner's decision to contract x_i is the only available instrument to affect the provision of the environmental service. In contrast to a perfect Pigouvian subsidy, the instrument x_i is a binding long-term contract, compliance may involve nonzero hassle costs (see, e.g., Jack and Jayachandran 2018), and, as is relevant in our auction setting, there may exist quantity constraints.¹

Landowner Types.—Each landowner i is characterized by a type $\theta_i = (c_i, a_{i0})$ distributed according to the cumulative distribution function $F(\theta)$. c_i is a landowner's cost of accepting the contract, or the minimum transfer p required for a landowner to accept the contract $x_i = 1$. c_i includes both a landowner's forgone option to crop over the contract period and any hassle costs associated with complying with the program rules.² a_{i0} is a landowner's action absent the contract. Supplemental Appendix A defines $\theta_i = (c_i, a_{i0})$ in a microfounded model.³

A landowner's expectations about the payoffs to cropping determine both the magnitude of the forgone option and the likelihood of cropping when unconstrained by the contract. One would therefore expect c_i and a_{i0} to be correlated

¹ A Pigouvian subsidy on all conserved land in the United States would be technologically and administratively challenging and would cost over 40 times the current budget of the CRP.

² Hassle costs are relevant across PES settings (Jack and Jayachandran 2018; White and Minang 2011). In the CRP specifically, contracts include a conservation plan developed with the Natural Resource Conservation Service, requirements to purchase specific seed mixes that vary in costs geographically, and mid-contract management requirements and possible audits, which may involve in-person interaction at a local USDA office. Indeed, free text survey responses among CRP participants cite a desire to reduce the amount of paperwork and recordkeeping mandated by the program (Allen and Vandever 2003).

³ In this model, each landowner is defined by a distribution of potential payoffs to cropping over the contract period and a hassle-cost type. Only the landowner's cost of accepting the contract c_i , which she reports to the mechanism, and her expected action absent the contract, which is unaffected by the mechanism, are relevant for landowner choices and social welfare (for the mechanisms we consider).

in the population, as landowners may have a low c_i in large part because they have unprofitable land that they do not plan to crop. But, landowners face uncertainty over the stochastic payoffs to cropping that determine their forgone option over ten-year contracts, and contracting in realistic settings involves hassle costs that enter c_i and may be arbitrarily correlated with a_{i0} . It will be useful to define the conditional expectation function:

$$(1) \quad \tau(c) = \mathbb{E}[1 - a_{i0} | c = c_i].$$

This function describes the expected additionality, or the expected impact of contracting on a_i , among all landowners with the same cost of accepting the contract.

Social versus Landowner Incentives.—The social surplus of contracting with landowner i is

$$(2) \quad SS_i = B \cdot (1 - a_{i0}) - c_i.$$

This is positive when the incremental value of environmental services due to contracting is higher than a landowner's costs of accepting that contract. Landowners choose $x_i = 1$ if $p \geq c_i$. Let

$$(3) \quad x_i^*(p) = \mathbf{1}\{p - c_i \geq 0\}$$

be landowner i 's choice to contract at price p . Equations (2) and (3) show that landowner i transacts based only on p and her cost c_i , but social surplus depends also on $1 - a_{i0}$, or her additionality. p will therefore not necessarily incentivize the highest social surplus landowners to contract.

Efficient Prices and Allocations.—The socially optimal uniform price solves⁴

$$(4) \quad \max_p \int (B \cdot \tau(c) - c) x^*(p; c) f_C(c) dc,$$

where the density f_C is the marginal of $F(\theta)$ on costs, c_i , and $x^*(p; c) = \mathbf{1}\{p - c \geq 0\}$. The solution to this problem is equivalent to one where a quantity is chosen and an allocation is implemented with a Vickrey auction.

Equation (4) shows that f_C and $\tau(c)$ are sufficient statistics for social welfare and landowner choices when p is the only instrument available to allocate landowners to contracts. More generally, f_C and $\tau(c)$ are sufficient statistics for social welfare for any incentive-compatible mechanism.⁵ Contracting with a landowner with cost c_i increases expected social surplus when

$$(5) \quad B \cdot \tau(c_i) - c_i \geq 0.$$

⁴ A uniform price and uniform B are motivated by the absence of observables. This could be because they have already been conditioned on, where equation (4) defines the pricing problem in a subpopulation.

⁵ See Lopomo, Persico, and Villa (2023) for more details and a proof and Einav, Finkelstein, and Cullen (2010) on the use of similar sufficient statistics for the analysis of adverse selection in competitive insurance markets.

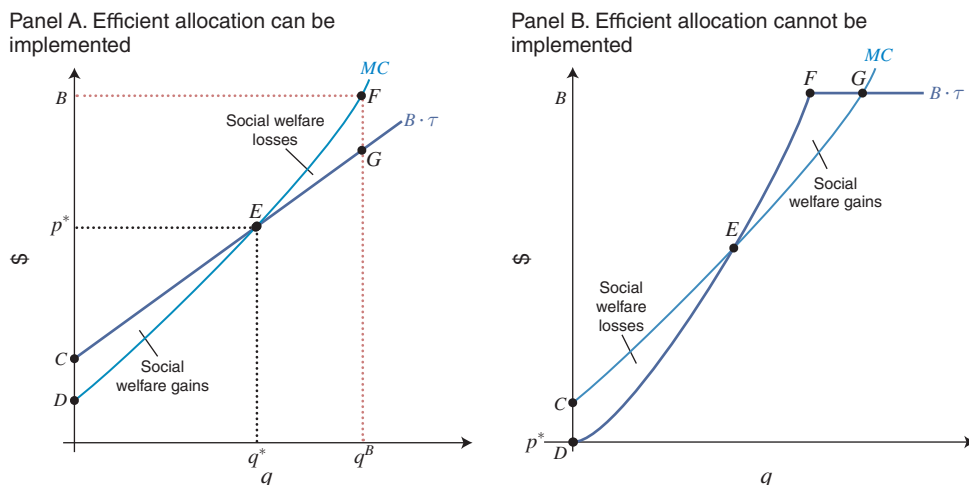


FIGURE 1. GRAPHICAL ANALYSIS

Notes: Figure describes markets characterized by marginal cost ($MC = F_C^{-1}(q)$) and contract value ($B \cdot \tau$) curves. The horizontal axis is the share of the population ordered by costs c_i . B denotes the social benefits of $a_{i1} = 1$. $B \cdot \tau$ denotes the incremental social benefits of contracting, relative to no contract, at each quantile of landowner costs. The vertical distance between the $B \cdot \tau$ and MC curves represents the social surplus from contracting. Upward-sloping $B \cdot \tau$ curves illustrate markets with adverse selection. Panel A documents a population distribution in which the efficient allocation (defined in equation (5)) can be implemented with the socially optimal uniform price p^* . In panel A, social surplus is monotonically decreasing in cost c_i . In panel B, social surplus is not monotonically decreasing in cost c_i . Panel B documents a population distribution in which the efficient allocation (defined in equation (5)) cannot be implemented. In panel B, the socially optimal price is $p^* = 0$ (no market). Panel A also demonstrates the social welfare losses from mis-pricing (at B) (triangle EFG).

Our interest in this stylized framework is in when an allocation that maximizes expected social surplus, $B \cdot \tau(c) - c$, is implementable. We will refer to this allocation as the efficient allocation.⁶

B. Graphical Analysis

We analyze the efficiency of allocations in the market graphically, plotting markets with different $F(\theta)$ in Figure 1, panel A and Figure 1, panel B. Each figure plots two curves: one based on f_C and one based on $\tau(c)$. The first curve is the inverse distribution function of costs c_i , $F_C^{-1}(q)$, or the *marginal cost (MC) curve*, where the horizontal axis q is the share of the population ranked by their costs of accepting the contract. The second curve is the value of contracting at each quantile of the distribution of c_i , $B \cdot \tau(F_C^{-1}(q))$, or the *contract value curve*.⁷ The contract value curve lies weakly below B , reflecting the possibility that $a_{i0} > 0$ for some landowners.

Each panel in Figure 1 displays an upward-sloping contract value curve ($\tau'(c) > 0$). This captures the fact that landowners' expectations about future payoffs

⁶We focus on this benchmark, following Lopomo, Persico, and Villa (2023), as it maximizes expected social surplus given cost reports c_i . This allocation achieves the maximum social welfare gain implementable by any mechanism; however, implementing this allocation will be possible only for some $F(\theta)$.

⁷We conduct the change of variables (to q) so that the areas between the contract value and marginal cost curves are interpretable as magnitudes of social welfare gains (or losses).

to cropping influence both c_i and a_{i0} —landowners who are likely to conserve may face a low cost of accepting a contract requiring conservation—and the existence of adverse selection in the market. Modeling adverse selection with an upward-sloping contract value curve builds on the widely used graphical analysis of adverse selection in insurance markets developed in Einav, Finkelstein, and Cullen (2010). The vertical distance between the contract value and marginal cost curves equals $B \cdot \tau(c) - c$, or the expected social surplus of contracting with all landowners with costs equal to c . From equation (5), it is efficient to contract only where the contract value curve lies above the marginal cost curve.

In Figure 1, panel A, the efficient allocation can be implemented with socially optimal incentives, satisfying $p^* = B \cdot \tau(p^*)$, at the intersection of the contract value and marginal cost curves. This implements social welfare gains in triangle CDE. Setting $p = B$, the social value of the conservation action, leads to inefficient contracting and social welfare losses (triangle EFG) where the incremental social benefits of conservation caused by the contract are less than the costs incurred among landowners.⁸

In Figure 1, panel B, the efficient allocation cannot be implemented. It is also socially optimal not to offer a market despite the existence of landowners for whom contracting is socially desirable. In Figure 1, panel B, the contract value curve lies below the marginal cost curve at low values of c_i (low q). This represents landowners that have low but positive costs of accepting a contract—due to some option value of cropping and/or hassle costs—but a high likelihood of conserving without the contract. In this market, a regulator cannot implement the efficient allocation (triangle EFG), as any incentive that is attractive for landowners in triangle EFG is also attractive for landowners in CDE, where losses in CDE outweigh gains in EFG.

The difference between equations (2) and (3) causes the inefficiency in Figure 1, panel B. The regulator can only affect allocations based on landowner costs, c_i , and the incentive p , but social surplus depends also on the impact of contracting, or a_{i0} . Because $B \cdot \tau(c) - c$, the vertical distance between the contract value and marginal cost curves, crosses zero from below in Figure 1, panel B, no mechanism can implement the efficient allocation (triangle EFG) as it would require an allocation rule that is not monotonically decreasing in landowner costs (see Myerson 1981; Lopomo, Persico, and Villa 2023).

The difference between Figure 1, panel A and Figure 1, panel B thus has implications beyond the illustrative-posted-prices mechanism in this section. In a procurement auction with a quantity constraint—our empirical setting—if $B \cdot \tau(c) - c$ is not decreasing in c , the efficient allocation subject to the quantity constraint may not be implementable by any mechanism. In Figure 1, panel A, $B \cdot \tau(c) - c$ is decreasing in c , but in Figure 1, panel B, it is not.

Figure 1 illustrates that social welfare and socially optimal incentives depend on f_C and $\tau(c)$. The goal of our empirical analysis is therefore to estimate them. However, this stylized model was limited in its tools; our empirical analysis will include a richer set of contracts and observable characteristics to investigate social welfare gains under various market designs.

⁸ Supplemental Appendix A discusses a special case of a one-dimensional model in which $p^* = B$ and empirical tests that distinguish this special case from the cases presented in Figure 1.

II. Setting and Data

A. *The CRP*

Our empirical setting is the CRP, a PES scheme incentivizing conservation on agricultural land administered by the USDA. Established in 1985, the CRP pays landowners between \$1.6 and \$1.8 billion per year to retire erodible and other environmentally sensitive cropland and undertake conservation actions through long-term contracts, typically ten years in duration.⁹ The CRP is one of the largest and most mature PES schemes in the world, and CRP contracts are similar to those in other PES programs (see Kinzig et al. (2011) and Salzman et al. (2018) for overviews).

Unlike the uniform pricing problem in Section I, the USDA awards CRP contracts via a complex auction mechanism. Under the CRP's general enrollment mechanism, eligible landowners bid for heterogeneous contracts in a discriminatory asymmetric-scoring auction.¹⁰ Contracts are differentiated by conservation actions that "top up" the base action of land retirement. These actions include planting specific grass mixes, planting or maintaining trees, and establishing or restoring pollinator or rare habitats.

Bids are scored according to a known scoring rule that awards bidders points for the environmental sensitivity of their land, their chosen contract, and their bid rental rate, a payment per acre per year. Characteristics of the land that determine points for environmental sensitivity include erodibility, importance for habitats, and potential for water and air pollution and carbon sequestration. Bid rental rates are subject to a bid cap based on the average land rental rate in the county and soil productivity estimates. Supplemental Appendix B describes the mechanism and scoring rule in more detail.

The aggregate acreage covered by CRP contracts is determined by Congress in the Farm Bill, which in turn determines the threshold score. All bidders with scores above the threshold score are awarded a contract.¹¹ Given uncertainty in both the aggregate acreage to be awarded contracts in that auction and their opposing bidders, bidders face uncertainty over the threshold score at the time of bidding. Bids are prepared with the assistance of staff at Farm Service Agency county offices, who help landowners trade off different contracts and bid rental rates.

Auctions occur every one to four years. Landowners are eligible to bid if they meet erosion standards, are in a national or state conservation priority area, and had cropped at least four years in a five- to ten-year window preceding the auction or are reenrolling CRP land.¹² The first two requirements target environmentally sensitive land. The final requirement is designed to limit participation to additional landowners. Landowners face steep penalties, refunding all payments since

⁹Most contracts are 10 years, but a small fraction of incentivized actions are associated with 15-year contracts.

¹⁰In addition to the general enrollment mechanism, there is a posted-price mechanism (continuous enrollment) for targeted land. Historically, approximately three quarters of CRP acreage has been contracted via the general enrollment mechanism (Hellerstein 2017).

¹¹There is also a constraint that no more than 25 percent of a county's land can be in the CRP, which almost never binds.

¹²Activities in the one to five years preceding bidding have no impact on CRP eligibility to eliminate any perverse incentives to crop land in order to become eligible or maintain eligibility for the CRP.

enrollment plus a 25 percent penalty, if they exit early or fail to comply with the rules of the program.

Research quantifying the value of the CRP has documented improvements in wildlife habitats, erosion control, water quality, and carbon sequestration from cropland retirement (Feather, Hellerstein, and Hansen 1999; Hansen 2007; FAPRI-MU 2007; Allen and Vandever 2012; Johnson et al. 2016; Hellerstein 2017). However, these analyses are typically conducted using models that ignore counterfactual land use. Perhaps motivated by eligibility requirements designed to restrict to additional landowners, the scoring rule is constructed under the assumption that land would be cropped in the absence of the program (Claassen, Horowitz, Duquette, and Ueda 2014).¹³ Because the primary motivation for the program and most environmental gains accrue from land retirement, relative to cropping, the possibility that some landowners conserve absent the CRP presents the additionality concern.

B. Data

Our dataset links bids to a panel of land use to measure additionality.

Data on Bids.—We obtain confidential administrative data on all components of the bid, including the bid rental rate, the bid contract and acreage, and the characteristics of landowners that impact the score. Our data cover all seven auctions that occurred from 2009 to 2021. We also obtain data on all CRP contracts (Farm Service Agency 2009–2021a,b).

Each landowner is defined by a collection of fields, or common land units, the smallest agricultural unit with a common land use. CRP contracts typically cover only a subset of a landowner's total fields. Our data include the geolocation of all bidding landowners for all auctions as well as identifiers for the specific fields offered into the mechanism ("bid fields") for one auction (in 2016). Auctions are nationwide, but bidders are concentrated in the Midwest, Southern Plains, and Southeast (see Supplemental Appendix Figure C.3a).

Data on Land Use.—We link bidders, and for the purpose of comparison, agricultural nonbidders, to a panel of land use outcomes. The primary land use outcome of interest is whether land is cropped versus retired, as this is the behavior incentivized by the CRP. We use both remote-sensing and administrative datasets due to their complementary strengths.¹⁴

Our primary dataset is the Cropland Data Layer (CDL), a remote-sensing product from the National Agricultural Statistics Service (NASS 2009–2020). This dataset provides land cover classifications at 30 meter (m) by 30 m resolution (roughly a quarter acre) from 2009 to 2020. The binary indicator of crop versus noncrop—our primary outcome of interest—is rarely misclassified (Lark, Schelly,

¹³This USDA report writes: "Benefit-cost indices are used to rank applications for acceptance in all major USDA conservation programs ... Existing indices, however, implicitly assume full additionality."

¹⁴The ability to conduct a geospatial linkage between micro-data on CRP bids and land use was made possible by recent efforts within the Economic Research Service (ERS) at the USDA. Previous micro-level analyses of land use among CRP participants used the National Resources Inventory (Roberts and Lubowski 2007). Contemporaneous work among ERS economists spearheaded linkages to administrative land use data ("Form 578") (Rosenberg and Pratt 2024), which facilitated a comprehensive linkage to remotely sensed land use outcomes.

and Gibbs 2021).¹⁵ However, as in other satellite-derived products, nonclassical measurement error can generate biases in assessing land use change (Alix-Garcia and Millimet 2022; Torchiana, Rosenbaum, Scott, and Souza-Rodrigues 2025).

Our second dataset is field-level administrative data on land use that agricultural landowners report to the USDA in Form 578 for 2013–2019 (Farm Service Agency 2013–2019). These data are accurate and comprehensive for cropped land because crop insurance payouts depend on them. However, landowners are only incentivized to report Form 578 if fields are insured by crop insurance, and landowners with CRP contracts are mechanically classified as noncropped. We also use a collection of high-resolution aerial imagery (1 m) of contracted land collected under the National Agriculture Imagery Program from 2017 to 2021 (Farm Production and Conservation Business Center 2017–2021). We use these images to observe and confirm compliance with CRP rules.

While accurate to assess agricultural land retirement—the main incentivized activity of the CRP—these datasets are much more limited in their ability to measure different “top-up” actions that differentiate the heterogeneous contracts in the mechanism (e.g., specific species). Our main estimates of additionality will focus on the measure that we can observe and the principal goal of the program: the binary outcome of retiring versus cropping land.

Supplemental Appendix C provides more details about agricultural land units, the construction of our dataset, the geographic distribution of cropland and CRP bidders (see Supplemental Appendix Figure C.3), and the use of aerial photographs to confirm compliance with CRP rules.

Summary Statistics.—Table 1 presents summary statistics. Columns 1 and 2 summarize all agricultural landowners in the United States, including CRP-eligible and ineligible landowners. Columns 3 and 4 summarize all land among bidders in our sample and columns 5 and 6 summarize bid fields.

Panel A presents land use outcomes in the year of bidding (prior to contract awards). Approximately 22–23 percent of all bidders’ land is cropped (20–24 percent on bid fields) versus 26–29 percent nationwide. The majority of the remainder is in natural vegetation and grassland. Corn and soybean cultivation account for two-thirds of all cropping. The remote-sensing and administrative data generally align.

Bidders have lower USDA-constructed estimates of soil productivity, are larger, and are more environmentally sensitive as measured by the scoring rule than the average agricultural landowner (panel B). These differences, along with the differences in land use in panel A, are likely driven in part by eligibility requirements that columns 1 and 2 are not conditioning on. The average bidder offers 84.1 acres into the CRP mechanism (33 percent of a bidder’s land) for a rental rate per acre per year of \$83. Two-thirds of bidders bid on a contract that includes a grassland action, 21 percent choose a wildlife habitat action, and 12 percent choose a tree-related action. 70 percent of bidders are recontracting after their initial contract expired.¹⁶ 81 percent of bidders are awarded contracts across the auctions in our sample. The average auction includes 36,772 bidders.

¹⁵The CDL has cropland superclass user and producer accuracy of over 95 percent from 2008 to 2016.

¹⁶Recontracting bidders are treated identically to new bidders by the scoring rule.

TABLE 1—SUMMARY STATISTICS

	All agricultural land		Bidders		Bid fields	
	Remote sensing (1)	Admin (2)	Remote sensing (3)	Admin (4)	Remote sensing (5)	Admin (6)
<i>Panel A. Land use</i>						
Share cropped	0.29	0.26	0.22	0.23	0.24	0.20
Share corn	0.11	0.10	0.08	0.08	0.07	0.06
Share soybean	0.09	0.09	0.07	0.08	0.09	0.09
Share fallow	0.02	0.01	0.03	0.02	0.04	0.02
Share nat. veg. or grassland	0.57		0.69		0.63	
<i>Panel B. Land characteristics</i>						
Size (acres)	160.7 (2,690.7)		250.6 (506.5)			
Soil productivity (\$/acre)	92.4 (63.2)		86.9 (58.5)			
Enviro. sensitivity (points)	53.5 (29.8)		86.5 (33.7)			
<i>Panel C. Bid characteristics</i>						
Rental rate (\$/acre/year)			83.0 (56.4)			
Acres bid			84.1 (136.3)			
Share recontracting			0.70			
Contract action = grasses			0.67			
Contract action = trees			0.12			
Contract action = habitat			0.21			
Share contracting			0.81			
Number of bidders/auction			36,772			
Observations	7,890,426		258,286		61,703	

Notes: Table presents summary statistics of all agricultural landowners (columns 1 and 2), bidding landowners (columns 3 and 4), and bid fields (columns 5 and 6), defined as the delineated land area entered into the mechanism to be awarded a CRP contract (observed only for bidders in the 2016 auction). Standard deviations in parentheses. Panel A reports land use outcomes in the remote-sensing (CDL) and admin (Form 578) data. All land use outcomes are reported for the year of the auction (prior to contract awards). Land use outcomes in columns 1 and 2 are averaged across auction years with land use outcomes among both eligible nonbidders and ineligible land. Land use categories follow Lark, Mueller, Johnson, and Gibbs (2017). Crop outcomes exclude alfalfa and hay. Soil productivity is calculated by NASS and is reported in dollars per acre. Environmental sensitivity points are the points given for characteristics of land in the scoring rule. Rental rate is reported in dollars per acre per year and is the dollar component of the bid in the auction. Acres bid is the total acreage entered into the auction to be awarded a CRP contract. Grasses, trees, and habitat contract indicators are aggregated over the menu of possible contracts.

III. Evidence: Additionality and Asymmetric Information

A. Regression Discontinuity Estimates of Additionality

We exploit the discontinuity at the winning-score threshold to estimate the treatment effect of a CRP contract in a regression discontinuity (RD) design. Our RD specification normalizes each landowner's score relative to her auction's win

threshold and evaluates how land use outcomes differ around this threshold. We estimate, for landowner, or bidder, i , in auction g , and year t ,

$$(6) \quad y_{igt} = \beta_{r(i,t)} \cdot \mathbf{1}\{S_{ig} \geq \underline{S}_g\} + f_{r(i,t)}(S_{ig} - \underline{S}_g) + \nu_{igt},$$

where S_{ig} is landowner i 's score in auction g , $\underline{S}_g$ is the winning-score threshold in auction g , $r(i,t) = t - t_{g(i)}$ normalizes the time dimension relative to the year of each auction ($t_{g(i)}$), and $f_{r(i,t)}(S_{ig} - \underline{S}_g)$ are relative-year-specific local-linear regressions in the mean squared error (MSE)-optimal bandwidth (Calonico, Cattaneo, and Titiunik 2014) allowed to differ for positive and negative values of $S_{ig} - \underline{S}_g$. $\beta_{r(i,t)}$ are time-varying RD coefficients relative to the year of the auction. We also estimate the pooled specification:

$$(7) \quad y_{igt} = \beta \cdot \mathbf{1}\{S_{ig} \geq \underline{S}_g\} + f(S_{ig} - \underline{S}_g) + \nu_{igt}.$$

Restricted to $r(i,t) > 0$, β is an estimate of the treatment effect at the margin of contract awards pooled over post-auction years. Treatment effect estimation (in equations (6) and (7)) therefore requires post-auction observations of land use outcomes, available for bidders in five auctions that occurred between 2010 and 2016.¹⁷ These auctions form the basis of our RD analysis sample, together with a panel of land use outcomes from 2009 to 2020 (remote-sensing data) and from 2013 to 2019 (administrative data). We estimate equations (6) and (7) at the landowner level to accommodate the possibility of spillovers to nonbid fields, and we cluster standard errors at the bidder level.

The RD design is valid under the assumption that bidders possess information about the winning-score threshold's distribution, but not its exact ex post realization. If bidders knew the threshold score, they would optimally bid just above it, invalidating the design. Figure 2, panel A presents a histogram of the score distribution normalized relative to the threshold score, $S_{ig} - \underline{S}_g$, the running variable of the RD, and confirms the lack of bunching at the winning-score threshold. Figure 2, panel B also shows no differential cropping at the discontinuity before the auction, plotting (binned) raw data and estimates of equation (7) for $r(i,t) \leq 0$.

Interpretation of equations (6) and (7) also requires an estimate of the magnitude of the first stage. Figure 2, panel C plots the share of bidders with a CRP contract after the auction around the award threshold (equation (7) for $r(i,t) > 0$) and demonstrates a first stage close to one. Figure 2, panel D plots the share of bidders' land covered by the contract (equation (7) for $r(i,t) > 0$), equal to approximately one-third among marginal winners.

Results.—Figure 3 presents data and estimates of equation (7) for $r(i,t) > 0$. As the CRP's primary goal is to incentivize agricultural land retirement, our outcome of interest is the share of each bidder's land that is cropped, which we measure in

¹⁷The last two auctions in our sample (in 2020 and 2021) have contract periods that begin after our panel of linked land use data concludes. Still, the variation from these auctions is useful to estimate our bidding model (Section IV).

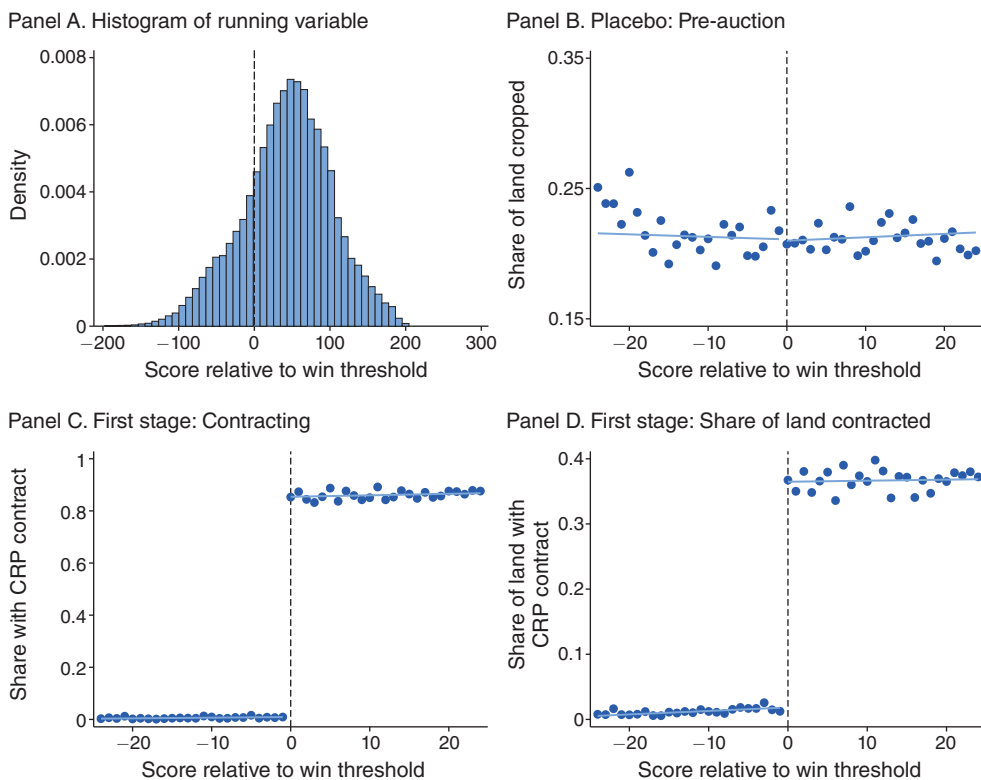


FIGURE 2. RD VALIDITY AND FIRST STAGE

Notes: Panel A presents a histogram of bidders' scores in the auction relative to that auction's win threshold, $S_{ig} - S_{g*}$, pooled across auctions in the RD sample. This is the running variable for the RD design: Bidders above zero win the auction. Panels B–D present raw data and estimates from equation (7). Panel B is estimated for $r(i, t) \leq 0$ (pre-auction), and panels C and D are estimated for $r(i, t) > 0$ (post-auction). The outcome in panel B is the share of the bidder's land that is cropped, measured in the remote-sensing data. The outcome in panel C is an indicator for a bidder obtaining a CRP contract, and the outcome in panel D is the share of a bidder's land covered by a CRP contract. Positive numbers on the x-axis correspond to winning scores; negative numbers correspond to losing scores. In panel A, each observation is a bidder; in panels B–D, each observation is a bidder year.

the remote-sensing data. Figure 3, panel A demonstrates that CRP contracts impact land use at the margin of contract awards. Landowners above the score threshold, who are awarded a contract, crop 8 percentage points less land than the marginal landowners who are not awarded contracts.¹⁸ This land is instead in natural vegetation and grassland (trees, shrubs, wetlands, and grasses), as incentivized by the CRP (Figure 3, panel B). The comparison between the magnitudes of land use impacts (Figure 3), relative to land covered and compensated under a CRP contract (Figure 2, panel D) forms the basis of our additional estimates. If contracting induced all marginal winners to change land use ($\tau = 1$), the magnitudes of the estimated treatment effects would align with the magnitudes of land covered by

¹⁸Cropping is not zero for winning bidders, as bidders typically only contract on a subset of their land.

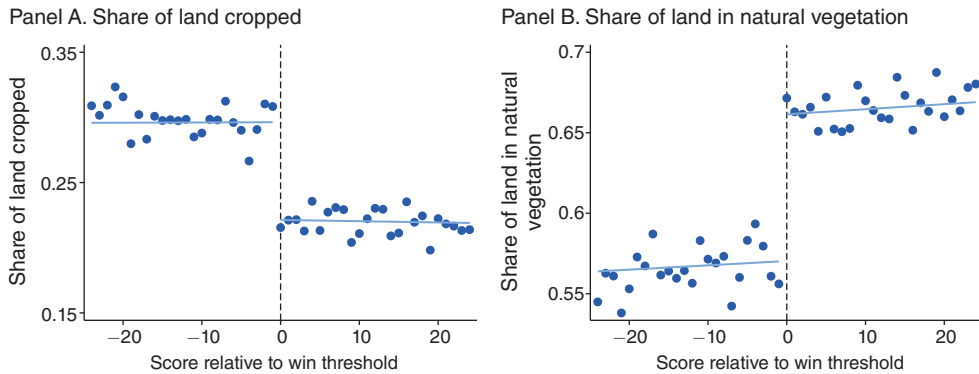


FIGURE 3. THE EFFECT OF A CRP CONTRACT ON LAND USE

Notes: Panels A and B present raw data and estimates from equation (7) for $r(i, t) > 0$ (post-auction). Outcomes are the share of the bidder's land that is cropped (A) and the share of the bidder's land that is in natural vegetation (trees, grassland, shrubs, and wetlands) (B), both measured in the remote-sensing data. The running variable is the difference between each bidder's score and the threshold score. Positive numbers on the x-axis correspond to winning scores; negative numbers correspond to losing scores. Each observation is a bidder year. Supplemental Appendix Figure D.3 provides corresponding figures with outcomes measured in the administrative data. Corresponding coefficient estimates and standard errors presented in Table 2.

the contract at the margin. The fact that they do not indicates that not all marginal winners are additional.

To analyze the time path of effects and illustrate our estimates of additionality, Figure 4 presents coefficient estimates of $\beta_{r(i, t)}$ from equation (6) and compares them to a 100 percent additional ($\tau = 1$) benchmark, estimated as the share of marginal bidders' land in a CRP contract (the pooled post-period estimate from equation (7)). We estimate equation (6) using both the remote-sensing data and the administrative data. Dashed lines represent pooled post-period estimates (equation (7)).

Figure 4 presents four facts. First, consistent with Figure 2, panel B, coefficient estimates are zero before the auction. Because the estimates in Figure 4 are relative-year-specific RD coefficients, pre-period effects are identified in levels. Second, post-period effect sizes and time trends are similar in both land use datasets, confirming that results are not driven by either nonclassical measurement error in the remote-sensing data or misreporting in the administrative data. Third, while treatment effects grow in the first two years as land transitions, effects are approximately constant over the ten-year contract period. Opportunities to rebid would attenuate treatment effects over time; we observe no evidence of decreasing magnitudes in Figure 4 and limited rebidding at the margin even after multiple opportunities (see Supplemental Appendix Figure D.4).¹⁹

Finally, Figure 4 demonstrates that over the ten-year period, the magnitude of the treatment effect of a CRP contract on land use is only approximately one-quarter of the $\tau = 1$ benchmark, or that approximately one-quarter of marginal winners

¹⁹ Even five years following the initial bid, only approximately 20 percent of losers have rebid (and fewer have won) despite multiple opportunities (see Supplemental Appendix Figure D.4, which plots the hazard rate of rebidding following a failed initial bid for all and marginal losers). This is consistent with the institutions: The general enrollment mechanism is shrinking with total acreage contracted decreasing on average over the sample period, limiting the option value of rebidding.

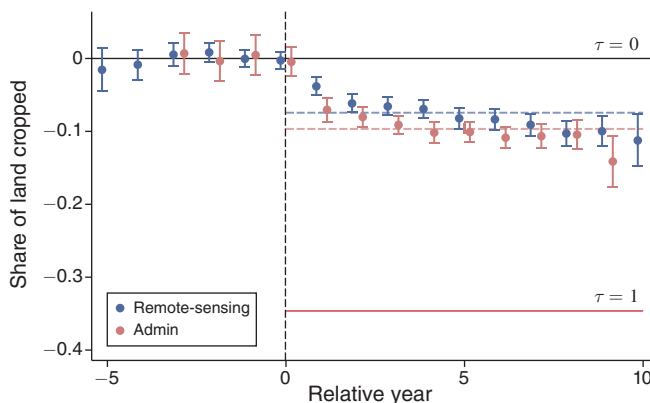


FIGURE 4. RD ESTIMATES OF ADDITIONALITY

Notes: Figure plots coefficient estimates from equation (6). The outcome is the share of each bidder's land that is cropped, measured with both remote-sensing and administrative datasets. The x -axis is the year relative to the year of each bidder's auction: $r(i, t) = t - t_{g(i)}$. Positive years correspond to post-auction years. Each point is an RD coefficient. Estimates pool data across all auctions with post-period land use data (occurring between 2010 and 2016); see Supplemental Appendix Table D.2 for auction-specific estimates. The relative-year coefficient estimates pool different auctions based on the coverage of the land use panel. Dashed lines indicate the pooled post-auction treatment effects (equation (7) estimated for $r(i, t) > 0$). The line labeled $\tau = 1$ represents a benchmark where all landowners are additional. This is calculated as the share of land contracting in the MSE-optimal bandwidth (Calonico, Cattaneo, and Titiunik 2014) used to estimate the RD. Each observation is a bidder year. Standard errors are clustered at the bidder level. Ten years is the full duration of a standard CRP contract. Corresponding coefficient estimates and standard errors presented in Table 2.

are additional. Table 2 presents point estimates from equations (6) and (7).²⁰ The main results in Table 2 quantify additionality estimates: Depending on the specification, we estimate rates of additionality at the margin of contract awards from 22 to 29 percent.

Estimates of additionality at the margin provide information about the $B \cdot \tau$ curve in Figure 1. First, it lies below B , as many landowners are not additional. Second, it highlights the need to estimate, rather than assume, the $\tau(c)$ function. If alternatively, costs and additionality could be summarized by a single index, in which bidders with positive costs are additional and bidders with costs equal to zero have additionality equal to zero, then at the margin additionality should be either zero or one (see Supplemental Appendix A); our estimates reject both of these hypotheses.²¹ Third, using variation in the winning-score threshold across auctions, we obtain estimates of additionality at the margin of 12 percent to 42 percent that are increasing in the quantile of the auction's marginal bidder (see Supplemental Appendix Table D.2). This cross-auction heterogeneity is consistent with an upward-sloping $B \cdot \tau$ curve.

²⁰ Supplemental Appendix Figures D.1, D.2, and D.3 present additional RD figures. Supplemental Appendix Tables D.3 and D.4 assess robustness to weighting each bidder by their bid CRP acreage (Supplemental Appendix Table D.3) and to restricting to bids of more than five acres (Supplemental Appendix Table D.4).

²¹ This interpretation is slightly complicated by bidder asymmetry, contract choices, and the pooling of auctions with different thresholds. We reestimate the RD specification in equation (7) split by the amount a bidder would need to bid for the base contract to achieve $\underline{S}_g$ and find $0 < \tau < 1$ across groups (results in Supplemental Appendix Table D.1).

TABLE 2—RD COEFFICIENT ESTIMATES

	Remote-sensing (1)	Admin (2)
<i>Panel A. Share of land cropped versus contracted</i>		
Share of land cropped pre-auction (placebo)	0.002 (0.007)	0.004 (0.011)
Share of land cropped post-auction (pooled)	−0.075 (0.007)	−0.097 (0.007)
Share of land cropped post-auction (average year effects)	−0.081 (0.007)	−0.101 (0.007)
Share of land contracted post-auction	0.345 (0.004)	
<i>Panel B. Other outcomes</i>		
Share of land cropped corn	−0.015 (0.003)	−0.022 (0.003)
Share of land cropped soybeans	−0.018 (0.003)	−0.025 (0.003)
Share of land fallow	−0.008 (0.002)	−0.012 (0.002)
Share of land in natural vegetation or grassland	0.091 (0.007)	
<i>Panel C. Spillovers to nonbid fields</i>		
Share of nonbid fields cropped	−0.007 (0.012)	−0.007 (0.011)
Observations in 25-point window		
Number of bidders	41,269	41,269
Number of bidder years	495,228	288,883
Spillovers (bid fields observed): Number of bidders	6,770	6,770
Spillovers (bid fields observed): Number of bidder years	81,240	47,390

Notes: Table presents coefficient estimates from equations (6) and (7) with land use outcomes measured in the remote-sensing (column 1) and administrative (column 2) data. All results use a local-linear regression on either side of the win threshold in the MSE-optimal bandwidth (Calonico, Cattaneo, and Titiunik 2014). All estimates are from equation (7), except the “average year effects” row, which presents the average of the post-period year effects (coefficient estimates of $\beta_{r(i,t)}$ from (6)) weighting each year equally. The pooled post-period includes an average of 7–8 post-auction years. Natural vegetation or grassland is only observed in remotely sensed data. Share of land contracted post-auction estimates the share of bidders’ land contracting among winning bidders. Panel C estimates the effect of a CRP contract on nonbid, and therefore noncontracting, fields to test for spillovers; this analysis is restricted to the 2016 auction. Standard errors are clustered at the bidder level. Estimates of additionality are calculated by dividing the estimates for the share of land cropped by the estimates of the share of land contracted and equal 22–23 percent (remote sensing) and 28–29 percent (admin). Estimates using the share of land in natural vegetation or grasslands yield a similar estimate (26 percent).

Beyond the RD.—As in the model in Section I, we argue that the estimates in Figure 4 are driven by heterogeneous land use absent the contract (a_{i0}) on the land bid into the mechanism. While in theory, both conservation without a CRP contract and cropping with a CRP contract (noncompliance) could contribute to the result in Figure 4, we find no evidence of noncompliance.²² Second, we document the absence of any spillovers onto nonbid fields (see panel C of Table 2).

²²We assess the CRP’s compliance regime by inspecting high resolution aerial photographs of over 1,000 contracted fields (see Supplemental Appendix C).

These results imply that we observe a realization of each bidder's a_{i0} on bid fields for all bidders who lose the auction. With knowledge that $a_{i1} = 1$, we observe a measurement of $1 - a_{i0}$ for each losing bidder. This provides rich information about heterogeneity beyond the RD for bidders below the threshold score.

B. Testing for Asymmetric Information

We use observations of landowners' realized additionality ($1 - a_{i0}$) and bids to conduct a test for asymmetric information in the spirit of Chiappori and Salanié (2000) and Hendricks and Porter (1988). We estimate the following specification:

$$(8) \quad 1 - a_{i0} = \beta \cdot \mathbf{b}_i + \pi \cdot \mathbf{z}_i + h(\mathbf{z}_i^s) + \nu_i,$$

where $1 - a_{i0}$ is measured as the share of landowner i 's bid fields that are cropped, observed only for rejected landowners without a contract award; $\mathbf{b}_i$ represents characteristics of i 's bid; $h(\mathbf{z}_i^s)$ are controls for characteristics that enter the scoring rule; and $\mathbf{z}_i$ are other landowner characteristics. We estimate multiple specifications of equation (8), but each one includes controls for the scoring rule, which impacts the strategic environment facing bidders. These include estimates of groundwater quality, surface water quality, wind and water erosion (deciles), air quality impacts, and whether a bidder is in a Wildlife Priority Zone or Air Quality Zone. We restrict to the one auction where we observe the delineations of bid fields (the 2016 auction); this is required to construct $1 - a_{i0}$. It is also the most restrictive auction, so $1 - a_{i0}$ is observed for 82 percent of bidders.²³ A positive correlation implies an upward-sloping contract value curve (see Section I), or adverse selection in the market.²⁴

Results.—We first document evidence of adverse selection. Figure 5, panel A presents a binned scatterplot of the correlation between additionality and the bid per acre-year (the bid rental rate), residualized of $h(\mathbf{z}_i^s)$. Figure 5, panel A demonstrates substantial heterogeneity in additionality and a systematic positive correlation between higher bids—reflecting higher costs of accepting a contract—and additionality. The interpretation of Figure 5, panel A is intuitive: Bidders with low costs of accepting a CRP contract have low costs in part because of private information that they would be likely to conserve without a CRP contract. Figure 5, panel B shows that bids remain correlated with additionality, capturing residual private information, conditional on other observables including prior land use interacted with estimates of the soil productivity of the bidders' land.

²³The low acceptance rate in 2016 resulted from an unusually small number of CRP acres available for enrollment, driven by the acreage cap set by the 2014 Farm Bill and a limited number of expiring contracts. Still, some bidders are awarded contracts, and in Section IV, we address the fact that equation (8) is estimated in a selected sample.

²⁴Relative to the approach to estimating adverse selection of Einav, Finkelstein, and Cullen (2010), our strategy leverages the fact that we observe *both* optimal bids *and* land use among rejected bidders. To compare the two approaches, it is useful to consider a simpler second-price auction for a single contract, where the distribution of costs therefore equals the distribution of bids. Then, because we can construct a measure of a_{i0} for each rejected bidder, $\tau(c)$ is simply the average $1 - a_{i0}$ at each value of the observed bid. With posted prices, as in Einav, Finkelstein, and Cullen (2010), estimating the distribution of costs (and $\tau(c)$) requires exogenous price variation to trace out this distribution.

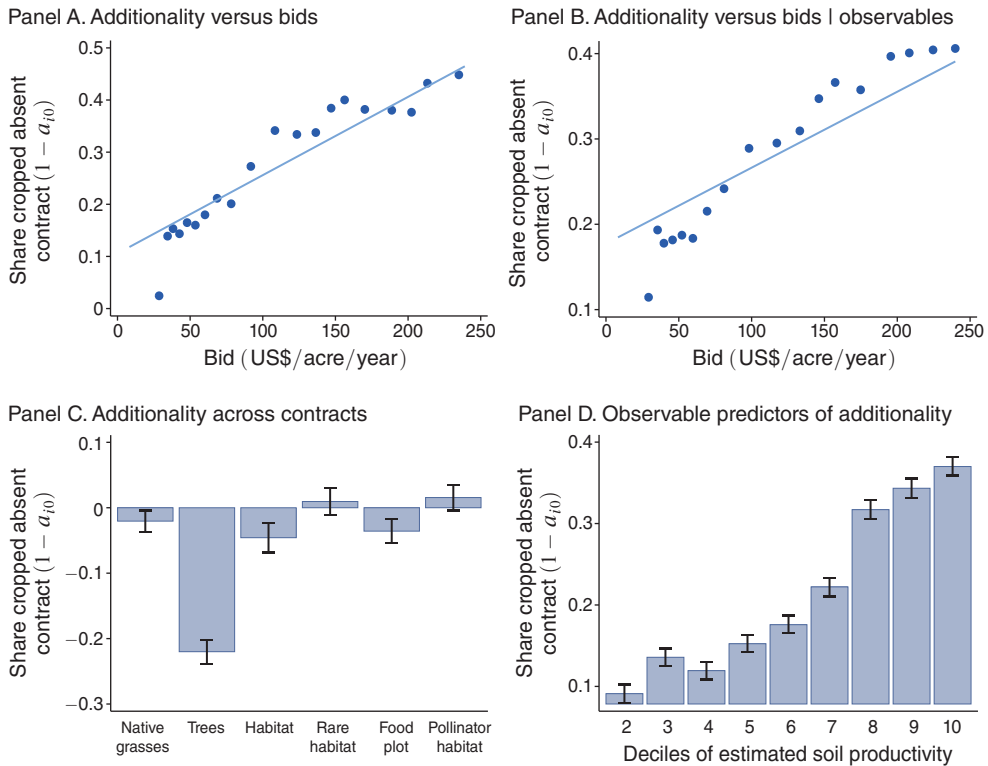


FIGURE 5. TESTING FOR ASYMMETRIC INFORMATION

Notes: Figure presents estimates from regression specifications in equation (8). Regressions in all panels control for landowner characteristics in the scoring rule: whether a bidder is in a Wildlife Priority Zone, estimates of groundwater quality, estimates of surface water quality, estimates of wind and water erosion (deciles), air quality impacts, and whether or not a bidder is in an Air Quality Zone. The outcome variable in all panels is a landowner-specific measure of additionality, calculated as the share of all fields bid into the CRP mechanism that are cropped post-auction for rejected landowners. The sample is restricted to the 2016 auction, in which 82 percent of bidders are rejected and the delineations of bid fields are observed. Cropping on bid fields is measured in 2017–2020 in the remote-sensing data (see Supplemental Appendix Figure D.5 for corresponding figures using the administrative data), with each field and year equally weighted. Panel A is a binned scatterplot correlating the dollar bid (per acre, per year) with additionality, conditional on characteristics included in the scoring rule. Panel B adds controls for interaction terms of prior land use (quartiles of share of land cropped prior to bidding and reenrolling CRP status) and deciles of estimated soil productivity. Panel C plots relative additionality by the chosen contract in the bid, relative to an omitted category of introduced grasses, conditional on characteristics included in the scoring rule. Panel D plots relative additionality by deciles of estimated soil productivity, conditional on characteristics included in the scoring rule. Standard errors clustered at the bidder level.

Next, we show that choices of contracts in the mechanism are systematically correlated with additionality. Figure 5, panel C replaces $\mathbf{b}_i$ in equation (8) with a vector of chosen contract indicators—the submitted bid on the menu of contracts—and documents substantial adverse selection (low additionality) on tree-related contracts, relative to the base category of introduced grasses. Figure 5, panel C highlights that contract choices reveal information about additionality and that alternative menus of contracts may lead to different outcomes in the market.

Finally, we present evidence that observable characteristics are predictive of additionality. Figure 5, panel D plots additionality by decile of landowner-predicted soil productivity, conditional on $h(\mathbf{z}_i^s)$ but excluding any endogenous bid choices

from the regression specification. These estimates of soil productivity are collected by the USDA and are designed to approximate the amount that a landowner would be able to earn on a given parcel of land. This characteristic is an ideal predictor of additionality in theory, and Figure 5, panel D demonstrates that it is predictive of additionality in practice. This result highlights the potential to differentiate incentives using this characteristic, which is not currently incorporated to rank bidders in the scoring rule.

Robustness.—The results in Figure 5 restrict to the 2016 auction, which is useful because we can construct a measure of $1 - a_{i0}$ and observe it for a large share of bidders below the threshold. Cross-auction heterogeneity suggests that the patterns in 2016 generalize across auctions. First, estimates that leverage heterogeneity in the location of the RD threshold, both pooling all auctions (Supplemental Appendix Table D.1) and exploiting cross-auction heterogeneity (Supplemental Appendix Table D.2), replicate both the range of additionality estimates—10–20 percent among lower bidders to about 40–50 percent among higher bidders—and the existence of adverse selection. Second, RD coefficients restricted to bids for tree-related contracts document null effects across all auctions (Supplemental Appendix Figure D.6). These two exercises illustrate the same patterns as our baseline analysis using both different auctions and different empirical strategies. Finally, while our baseline approach hinges on delineating bid fields (only possible in 2016), we examine whether tract-level heterogeneity in cropping in the year of the auction follows stable patterns of heterogeneity. Tract-level cropping rates demonstrate similar, albeit not identical, heterogeneity across observables across auctions (Supplemental Appendix Figure D.7).

IV. Empirical Model of Bidding and Additionality

To quantify the welfare implications of the results in the preceding section, we develop a model of bidding and additionality to estimate (i) the distribution of landowner costs among a menu of contracts and (ii) additionality as a function of costs and observable characteristics. Together with estimates of the social benefits of CRP actions, the model facilitates analysis of social welfare under current and counterfactual market designs.

Landowners with a vector of private costs bid on discrete contracts, differentiated by heterogeneous conservation actions, in response to a nonlinear scoring rule. Landowners also differ in their additionality, which we model with a conditional expectation function that depends on landowners' characteristics and costs. Our empirical strategy first uses the optimality of bidding in the auction to estimate bidders' costs by revealed preferences and then estimates expected additionality as a function of costs and landowner characteristics by matching moments of the observed joint distribution of land use, landowner characteristics, and optimal bids.

A. Model

There are N landowners, indexed by i , and J contracts, indexed by j . Each landowner is characterized by (i) her costs (c_i, κ_i) for $\kappa_i = \{\kappa_{ij}\}$ and (ii) her land

retirement conservation action a_{i0} absent the CRP. Extending the model in Section I, redefine $F(\theta)$ as the cumulative distribution function of landowner types $\theta_i = (c_i, \kappa_i), a_{i0}$ and $F_{c,\kappa}$ as the marginal on (c_i, κ_i) .

The vector (c_i, κ_i) defines landowner i 's costs of accepting a contract. A landowner's cost of contract j is $c_i + \kappa_{ij}$, where c_i is the base cost of accepting the contract, common across contracts, and κ_{ij} is the top-up cost associated with contract j . We assume (c_i, κ_i) are drawn independently across landowners conditional on observables $\mathbf{z}_i$.

It will again be useful to define the function:

$$(9) \quad \tau(\mathbf{z}_i, c_i, \kappa_i) = \mathbb{E}[1 - a_{i0} | \mathbf{z}_i, c_i, \kappa_i]$$

or the expected difference in land retirement (conservation) with a CRP contract versus without a CRP contract given observable characteristics $\mathbf{z}_i$ and landowner costs (c_i, κ_i) .

Auction.—Landowners (bidders) submit a two-part bid $\mathbf{b}_i = (r_i, \mathbf{x}_i)$. $\mathbf{x}_i$ is a contract vector, with $x_{ij} = 1$ if the j -th contract is chosen and $x_{ij} = 0$ otherwise. Landowners choose a single contract, so $\sum_j x_{ij} = 1$. If landowner i wins the auction, $\mathbf{b}_i$ describes the terms of her CRP contract: She performs the action defined in $\mathbf{x}_i$ and receives a payment of r_i dollars per acre-year. Each bid $\mathbf{b}_i$ is converted into a score according to a known scoring rule $s(\mathbf{b}_i, \mathbf{z}_i^s)$. All landowners above a winning-threshold score $\underline{S}$ are awarded a contract.

Landowner i forms beliefs about the probability of winning the auction with a score S given uncertainty over her competitors and the acreage limit of the auction. We assume that landowner i does not observe the number or characteristics of her competitors. All landowners form the same beliefs about the distribution of the threshold score $\underline{S}$. Define $G(S) = \Pr\{S \geq \underline{S}\}$.²⁵

Each landowner i solves

$$(10) \quad \mathbf{b}_i^* = \arg \max_{(r, \mathbf{x})} \left\{ \underbrace{(r - c_i - \mathbf{x} \cdot \kappa_i)}_{\text{Payoff to } i \text{ conditional on bid } (r, \mathbf{x})} \times \underbrace{G(s((r, \mathbf{x}), \mathbf{z}_i^s))}_{\text{Probability of } i \text{ winning with bid } (r, \mathbf{x})} \right\},$$

where a landowner chooses her bid $\mathbf{b}_i = (r_i, \mathbf{x}_i)$ to maximize her payoff conditional on winning, multiplied by the probability of winning with that bid, given her costs (c_i, κ_i) . Landowners compete on both r and $\mathbf{x}$ in equation (10), but whether a landowner wins against her competitors depends only on her score. A landowner's best response in equation (10) therefore solves both an "inner problem" and an "outer problem," building on Asker and Cantillon (2008) and Che (1993).

²⁵ Beliefs $G(S)$ integrate over the strategies of the other bidders, the number of competing bidders, and the aggregate acreage limit. This approach is motivated by the fact that bidding is decentralized among thousands of bidders across the United States, and we document evidence of quantity uncertainty empirically (see Supplemental Appendix Figure E.1). We assume that beliefs are consistent with the empirical distribution of scores, following standard approaches to auctions (Hortaçsu 2000; Hortaçsu and McAdams 2010) and other incomplete information games (Agarwal and Somaini 2018; Agarwal, Li, and Somaini 2023).

The landowner's inner problem solves for the optimal payment and contract $(r(S, \mathbf{x}_i^*(S), \mathbf{z}_i^s), \mathbf{x}_i^*(S))$ to reach a target score S , where $r(S, \mathbf{x}, \mathbf{z}_i^s)$ is the bid payment required to obtain score S with contract $\mathbf{x}$, implicitly defined by $s((r(S, \mathbf{x}, \mathbf{z}_i^s), \mathbf{x}), \mathbf{z}_i^s) = S$. For any target score S , a landowner's best response must satisfy

$$(11) \quad r(S, \mathbf{x}_i^*(S), \mathbf{z}_i^s) - \mathbf{x}_i^*(S) \cdot \kappa_i \geq r(S, \mathbf{x}, \mathbf{z}_i^s) - \mathbf{x} \cdot \kappa_i,$$

for all $\mathbf{x}$ where $s((r(S, \mathbf{x}, \mathbf{z}_i^s), \mathbf{x}), \mathbf{z}_i^s) = S$. $r(S, \mathbf{x}, \mathbf{z}_i^s)$ defines the menu of payments awarded for each contract under the scoring rule (see Supplemental Appendix Table B.3 for an example) and determines the incentives facing landowners in the inner problem. Because the scoring rule is decreasing in r , $r(S, \mathbf{x}, \mathbf{z}_i^s)$ is increasing in the points awarded to $\mathbf{x}$, raising the probability that $\mathbf{x}$ is a landowner's optimal choice of contract, all else equal, for any target score S . Similarly, from equation (11), all else equal, a landowner with a lower κ_{ij} will be more likely to choose a contract with $x_{ij} = 1$.

Given (11), the best response $\mathbf{b}_i^* = (r(S_i^*, \mathbf{x}_i^*(S_i^*), \mathbf{z}_i^s), \mathbf{x}_i^*(S_i^*))$ in equation (10) must also satisfy

$$(12) \quad \begin{aligned} G(S_i^*) & (r(S_i^*, \mathbf{x}_i^*(S_i^*), \mathbf{z}_i^s) - c_i - \mathbf{x}_i^*(S_i^*) \cdot \kappa_i) \\ & \geq G(S) (r(S, \mathbf{x}_i^*(S), \mathbf{z}_i^s) - c_i - \mathbf{x}_i^*(S) \cdot \kappa_i) \end{aligned}$$

for all integer scores S . This is the landowner's outer problem solving for her optimal score S_i^* , trading off changes in the probability of winning the contract with her payoff conditional on winning. All else equal, it will be optimal for higher-cost landowners to choose lower scores.²⁶

Additionality.—In the contract period, landowners make land use decisions. If awarded a contract, $a_{i1} = 1$. If not, landowners choose a_{i0} , which is not contractible. At the time of bidding, equation (9) is the population expectation of $1 - a_{i0}$ for landowners with observable characteristics $\mathbf{z}_i$ and costs (c_i, κ_i) . We estimate the function $\tau(\mathbf{z}_i, c_i, \kappa_i)$ instead of modeling the choice of a_{i0} because together with $F_{c, \kappa}$, this conditional expectation function is sufficient to calculate social welfare under counterfactuals that hold the menu of contracts fixed.

Although the mechanism is more complex, the source of inefficiency is the same as in Section I: Landowner choices and allocations depend only on the scoring rule $s(\mathbf{b}_i, \mathbf{z}_i^s)$ and costs (c_i, κ_i) , but the efficient allocation depends also on $\tau(\mathbf{z}_i, c_i, \kappa_i)$.

B. Identification and Estimation

We assume that beliefs $G(S)$ and the scoring rule, which defines $r(S, \mathbf{x}, \mathbf{z}_i^s)$, are known or can be estimated. Then, the inequalities in equations (11) and (12) defining

²⁶Consider a one-point upward deviation from S_i^* to $S_i^* + 1$ when $\mathbf{x}_i^*(S_i^*) = \mathbf{x}_i^*(S_i^* + 1)$. Then, from equation (12), for S_i^* to be optimal, it must be the case that

$$c_i + \mathbf{x}_i^*(S_i^*) \cdot \kappa_i \geq r(S_i^*, \mathbf{x}_i^*(S_i^*), \mathbf{z}_i^s) + (r(S_i^* + 1, \mathbf{x}_i^*(S_i^* + 1), \mathbf{z}_i^s) - r(S_i^*, \mathbf{x}_i^*(S_i^*), \mathbf{z}_i^s)) \frac{G(S_i^* + 1)}{G(S_i^* + 1) - G(S_i^*)}.$$

Similarly, all else equal, lower-cost landowners will choose higher scores (consider instead a one-point downward deviation from S_i^* to $S_i^* - 1$).

a landowner's best response link the observed bids $\mathbf{b}_i^* = \left(r(S_i^*, \mathbf{x}_i^*(S_i^*), \mathbf{z}_i^s), \mathbf{x}_i^*(S_i^*) \right)$ and underlying costs (c_i, κ_i) and incorporate all restrictions on preferences given the observed bids.²⁷ These optimality conditions defining the bidders' best response is the basis for our identification argument and empirical approach, following Agarwal, Li, and Somaini (2023). It is also clear from these equations that without further variation in the mechanism or parametric assumptions, the vector of costs (c_i, κ_i) for each individual bidder cannot be point identified from the observed bids: The vector of costs has dimension J , while the optimal score is determined only for the single optimal contract $\mathbf{x}_i^*(S_i^*)$. In principle, with sufficient exogenous variation in the scoring rule, one could "trace out" the distribution of (c_i, κ_i) conditional on observable characteristics $\mathbf{z}_i$, $F_{c, \kappa | \mathbf{z}}$, using the best response inequalities in equations (11) and (12) (Agarwal, Li, and Somaini 2023). Tracing out $F_{c, \kappa | \mathbf{z}}$ without any parametric assumptions requires full-support variation in the vector of payments $r(S, \mathbf{x}, \mathbf{z}_i^s)$ for each contract $\mathbf{x}$ across target scores S .²⁸ This is a demanding requirement.

Identification of $\tau(\mathbf{z}_i, c_i, \kappa_i)$ extends this logic. It relies on the fact that we observe the joint distribution of optimal bids *and* land use for rejected bidders. If we were to observe an auction where bidders submitted bids given beliefs $G(S)$, but all bidders were unexpectedly and exogenously rejected, then counterfactual land use (a_{i0}) would be observed for the entire population. $\tau(\mathbf{z}_i, c_i, \kappa_i)$ would be identified based on the argument above: $\mathbb{E}[1 - a_{i0} | \mathbf{z}_i, \mathbf{b}_i^*]$ is observed and the revealed preference inequalities in equations (11) and (12) link $\mathbf{b}_i^*$ to costs (c_i, κ_i) . However, even in the auction with the most rejected bidders in our sample, a_{i0} is *not* observed for the 18 percent of bidders who are awarded a contract. Without this ideal "cancelled auction" experiment, variation in the mechanism to replicate it, or parametric assumptions, the entire function $\tau(\mathbf{z}_i, c_i, \kappa_i)$ cannot be learned from the rejected bidders alone. Restricting to the observed thresholds in the data, variation in the scoring rule across bidders can replicate the "cancelled auction" if scores are sufficiently low for some values of $\mathbf{z}_i^s$ independent of (c_i, κ_i) and a_{i0} . Or, by parameterizing $\tau(\mathbf{z}_i, c_i, \kappa_i)$, we can extrapolate based on the observations of bids *and* land use among rejected bidders.

Our approach relies on a combination of exogenous scoring-rule variation—both cross-auction and within-auction cross-landowner—and parametric extrapolation to address the limits to the variation we observe in the data. Some sources of variation differentially affect certain contracts, shifting relative incentives in the inner problem in equation (11); others shift the level of the scoring rule for some bidders but not others.

The first source of scoring-rule variation is a mid-mechanism policy change: After bids were initially collected in 2021, additional payments based on contracts' carbon sequestration potential were announced and bids were re-collected under this new scoring rule. Under the final mechanism, a landowner could increase the bid to obtain the same target score $r(S, \mathbf{x}, \mathbf{z}_i^s)$ by either 3 percent, 5 percent, or 10 percent, relative to the interim rule, depending on the chosen contract. We obtained the bids submitted

²⁷ Intuitively, the inner problem defines inequalities for κ_i based on the relative payments for chosen *versus* nonchosen contracts at the optimal score S^* , and the outer-problem inequality provides bounds on $c_i + \mathbf{x}_i^*(S_i^*) \cdot \kappa_i$.

²⁸ Define $\mathbf{r}$ as the vector of payments to obtain a target score S for each contract, $\mathbf{r} = \{r(S, \mathbf{x}^j, \mathbf{z}_i^s)\}_1^J$ (where $\mathbf{x}^j$ denotes the contract with $x_{ij} = 1$). The argument requires full-support variation in $\mathbf{r}$ to trace the tails of $F_{c, \kappa | \mathbf{z}}$. For intuition, consider increasing relative payments $r(S, \mathbf{x}^j, \mathbf{z}_i^s) - r(S, \mathbf{x}^{j'}, \mathbf{z}_i^s)$ for all S and all j' . From equation (11), as $r(S, \mathbf{x}^j, \mathbf{z}_i^s) - r(S, \mathbf{x}^{j'}, \mathbf{z}_i^s)$ increases for all j' , an increasing number of bidders will choose $\mathbf{x}_i^j$, ultimately replicating the familiar case of an auction for a single contract, $\mathbf{x}^j$.

in both the interim and final mechanism, providing two different functions $r(S, \mathbf{x}, \mathbf{z}_i^s)$ across the two auctions with the same bidders and contract period. In a second source of variation, before 2011, bidders in Wildlife Priority Zones (WPZs) received extra points regardless of contract $\mathbf{x}$; after, they were awarded 30 WPZ points only if they chose a subset of contracts, providing variation in $r(S, \mathbf{x}, \mathbf{z}_i^s)$ across bidders and auctions. Finally, whether or not a bidder is in an Air Quality Zone (AQZ) shifts the level of the score. Supplemental Appendix Figure E.2 summarizes the scoring-rule variation by plotting $r(S, \mathbf{x}, \mathbf{z}_i^s)$ for each contract $\mathbf{x}$ across the changes described above, and Supplemental Appendix Figure E.3 provides more details on individual changes.²⁹

The scoring-rule variation is useful but limited in practice, both in the differential incentives across contracts and in replicating the “canceled auction” thought experiment. We therefore parameterize $F_{c, \kappa | \mathbf{z}}$ and $\tau(\mathbf{z}_i, c_i, \kappa_i)$. We parameterize (c_i, κ_i) as

$$(13) \quad c_i \sim N(c(\mathbf{z}_i), \sigma_c^2(\mathbf{z}_i)) \quad \kappa_{ij} = p_j(\mathbf{z}_i) + u_j(\mathbf{z}_i) + \epsilon_{ij} \quad \epsilon_{ij} \stackrel{i.i.d.}{\sim} N(0, \sigma_\kappa^2(\mathbf{z}_i)),$$

with c_i and κ_{ij} drawn from independent normal distributions with means and variances that depend on observable characteristics, $\mathbf{z}_i$. κ_{ij} depends on contract features: p_j defines mean costs for a vector of primary covers, relative to the base category of introduced grasses (normalized to zero), and u_j is a vector of upgrade covers plus the no-upgrade option (normalized to zero).³⁰ We also parameterize

$$(14) \quad \tau(\mathbf{z}_i, c_i, \kappa_i) = \pi \cdot \mathbf{z}_i + \beta \cdot c_i + \alpha \cdot \kappa_i.$$

This specification allows $\tau(\mathbf{z}_i, c_i, \kappa_i)$ to depend on observable characteristics, $\mathbf{z}_i$, and unobserved bidder costs (c_i, κ_i) , where we align the dimension of α with the primary and upgrade parameterization of κ_{ij} —that is, we impose that $\alpha_j = \alpha_{j'}$ if $p_j = p_{j'}$ and $u_j = u_{j'}$.

Estimation.—Estimation proceeds in three steps. Supplemental Appendix E provides details.

Step 1 (Simulate $G(S)$): We estimate beliefs $G(S)$ with a resampling approach following Hortaçsu (2000); and Hortaçsu and McAdams (2010). First, we fit beta distributions to the number of bidders and acreage limits across auctions. We use additional historic data on acreage limits and the numbers of bidders for all auctions from 2000 to 2021 to fit these distributions. Then, we simulate the numbers of

²⁹To contextualize the magnitude of the variation in the scoring rule, it is helpful to compare it to the variation in the threshold score. The variation in $r(S, \mathbf{x}, \mathbf{z}_i^s)$ is of reasonable but modest magnitude: The WPZ variation yields a change in $r(S, \mathbf{x}, \mathbf{z}_i^s)$ of approximately \$8 for affected contracts. The mid-mechanism policy change affects different contracts, but the magnitude of the changes (in $r(S, \mathbf{x}, \mathbf{z}_i^s)$) is similar. By comparison, the difference between $r(S, \mathbf{x}, \mathbf{z}_i^s)$ for the base contract at the average-threshold score versus the 2016-threshold score is over \$20. See Supplemental Appendix Figure E.2 for details. A final important limitation is that these sources of variation shift relative incentives across groups of contracts: Even combining all of the sources of variation, $r(S, \mathbf{x}^j, \mathbf{z}_i^s) - r(S, \mathbf{x}^{j'}, \mathbf{z}_i^s)$ for some comparisons remains constant.

³⁰Landowners face a discrete choice over each of the primary and upgrade covers, but primary and upgrade covers can be combined. There are 36 total possible contracts, reflecting finer categorizations beyond the five dimensions in p_j (12 total) that each can be combined with an upgrade option. See Supplemental Appendix B for more details. In principle, costs may vary or be correlated across each of the 36 contracts in unmodeled ways; this parameterization attempts to parsimoniously capture key differences across contracts.

bidders and the acreage thresholds and resample from the observed joint distribution of the scores and acreages of bidders within each simulated auction. $G(S)$ is the share of simulated auctions in which S is a winning score.

Step 2 (Estimate Costs): Given estimates of beliefs $G(S)$ and the known scoring rule, the next step estimates $F_{c,\kappa|\mathbf{z}}$ using the parameterization in (13) and the optimality of observed bids as the solution to equation (10) via maximum simulated likelihood. The likelihood of each observed optimal bid, given $F_{c,\kappa|\mathbf{z}}$, is the probability that equations (11) and (12) are satisfied at $\mathbf{b}_i^* = (r(S_i^*, \mathbf{x}_i^*(S_i^*), \mathbf{z}_i^*), \mathbf{x}_i^*(S_i^*))$, which we obtain by checking all feasible deviations for each bidder across simulation draws of (c_i, κ_i) from a candidate $F_{c,\kappa|\mathbf{z}}$. Because the choice set of feasible bids is large, we coarsen the observed bid space used to construct each bidder's likelihood contribution in estimation.³¹ We classify bidders into 32 categories of observable types $\mathbf{z}_i$ that parameterize $F_{c,\kappa|\mathbf{z}}$ based on interactions of quartiles of soil productivity, prior CRP status, and prior land use status. We estimate $F_{c,\kappa|\mathbf{z}}$ using all auctions in our sample, including both the interim and final mechanism in 2021, which incorporate all the scoring-rule variations described above.

Step 3 (Estimate Additionality): The third step estimates the parameters in equation (14), (π, β, α) , via the method of simulated moments (MSM). This estimator searches for the parameters (π, β, α) that rationalize moments of the joint distribution of $1 - a_{i0}$, $\mathbf{b}_i^*$, and $\mathbf{z}_i$ observed in the data. Specifically, we simulate (c_i^k, κ_i^k) from $F_{c,\kappa|\mathbf{z}}$ estimated in step 2, solve for the optimal $\mathbf{b}_i^{*k}$ for each simulation draw k , and search for the parameters (π, β, α) that match (i) additionality at the winning-score threshold, (ii) additionality among all rejected bidders and by observable characteristics, (iii) the covariance between additionality and chosen scores, and (iv) the additionality among all landowners choosing a given contract.

We estimate $\tau(\mathbf{z}_i, c_i, \kappa_i)$ using rejected bidders from the 2016 auction; this is the auction with the most rejected bidders in which we observe outcomes on bid fields for the 82 percent of bidders that are rejected (as in Figure 5). We assume that the function $\tau(\mathbf{z}_i, c_i, \kappa_i)$ is stable across years, so the auction with the most rejected bidders provides the most information about counterfactual land use.³² This auction incorporates the cross-landowner variation in WPZ and AQZ status, which shifts scores; however, even with this variation, we must rely on parametric extrapolation for the highest-scoring bidders.³³ We measure additionality as the share of bid fields

³¹ We coarsen the observed choice of score into deciles of the score distribution and the choice of contract into seven categories corresponding to the seven dimensions of p_j and u_j . We do not coarsen the choice set when solving the bidder's problem. We use importance sampling (following Akerberg 2009) to reduce computational burden.

³² In Supplemental Appendix G, we assess robustness to the assumption of stability of $\tau(\mathbf{z}_i, c_i, \kappa_i)$ and our baseline estimation approach that relies exclusively on land use data from the 2016 auction. We extend our analysis of counterfactual auctions to construct scoring rules based instead on additionality estimates that match cross-tract heterogeneity in cropping in the year of the auction (Supplemental Appendix Figure D.7) and cross-auction heterogeneity in RD coefficients (Supplemental Appendix Table D.2). This approach allows us to probe the assumption of stability of $\tau(\mathbf{z}_i, c_i, \kappa_i)$ by constructing a scoring rule in a sample of (previous) auctions, comparing it to our baseline rule (Supplemental Appendix Figure G.5), and assessing performance in a holdout auction (Supplemental Appendix Figure G.6).

³³ Because the variation in $r(S, \mathbf{x}, \mathbf{z}_i^*)$ across changes to the scoring rule is relatively small compared to the differences in bid payments at different score thresholds (see Supplemental Appendix Figure E.2), in the most restrictive 2016 auction, we observe a measurement of a_{i0} for many landowners whose optimal bid would win in other auctions across any observed variations to the scoring rule. This motivates our estimation approach.

TABLE 3—MEAN LANDOWNER COSTS OF ACCEPTING CRP CONTRACTS

	All (1)	Bidders with > median soil productivity (2)
Base cost (c_i)	67.21	92.86
Top-up cost (κ_{ij})		
Native grasses	−0.03	3.97
Trees	23.76	27.33
Habitat	15.06	18.61
Rare habitat	15.44	18.91
Wildlife food plot	18.79	15.09
Pollinator habitat	17.97	17.41

Notes: Table presents estimated mean landowner costs for the base cost c_i and top-up cost κ_{ij} in dollars per acre per year. The base contract of introduced grasses is normalized to zero. The cost of each contract j is defined as $c_i + \kappa_{ij}$. Costs are estimated using revealed preferences in optimal bidding (equation (10)). Column 1 presents mean costs for all bidders across all auctions, and column 2 restricts to bidders with above-median soil productivity.

that are cropped over the contract period in the remote-sensing data, where we weight each field and year from 2017 to 2020 equally. The observables $\mathbf{z}_i$ in $\tau(\mathbf{z}_i, c_i, \kappa_i)$ are the 32 observable bidder types that parameterize $F_{c, \kappa | \mathbf{z}}$ and the components of the scoring rule excluding WPZ and AQZ status. We condition on optimal bids that are below the score threshold.

We conduct robustness checks to evaluate the sensitivity of results to parametric assumptions in Supplemental Appendix E. First, we estimate alternative functional forms of $\tau(\mathbf{z}_i, c_i, \kappa_i)$. Second, we conduct an exercise to examine how additionality varies with costs without parametric assumptions. To do so, we conduct an inversion exercise in the spirit of Guerre, Perrigne, and Vuong (2000), holding fixed bidders' choice $\mathbf{x}_i$ and estimate additionality as the average cropping rates among rejected bidders across quantiles of this inverted value. We emphasize that this exercise is not directly interpretable, as it selects on bidders' endogenous choice of contract in response to the nonlinear scoring rule. Nevertheless, it is useful to assess that our conclusions are not driven by our assumed functional forms.

C. Estimates

Table 3 summarizes mean costs across contracts, and Figure 6, panel A plots the estimated distribution of c_i .³⁴ A large share of landowners have low values of c_i , below \$50 per acre per year with a tail of bidders with higher c_i . Relative costs, with the base contract of introduced grasses normalized to zero, are generally intuitive. Higher-soil-productivity bidders have higher costs for primary covers but lower costs for upgrade covers. Because costs are estimated using only revealed preferences in bids, we examine whether estimated costs correlate with land use. Figure 6, panel B presents a binned scatterplot of $1 - a_{i0}$ against simulated costs c_i (conditional on observables) for rejected bidders. Figure 6, panel B documents that

³⁴ Supplemental Appendix Table E.2 presents parameter estimates for select cells of $\mathbf{z}_i$ and standard errors. Supplemental Appendix Table E.1 reports results estimated using beliefs based on the distribution of scores in the previous (versus the current) auction.

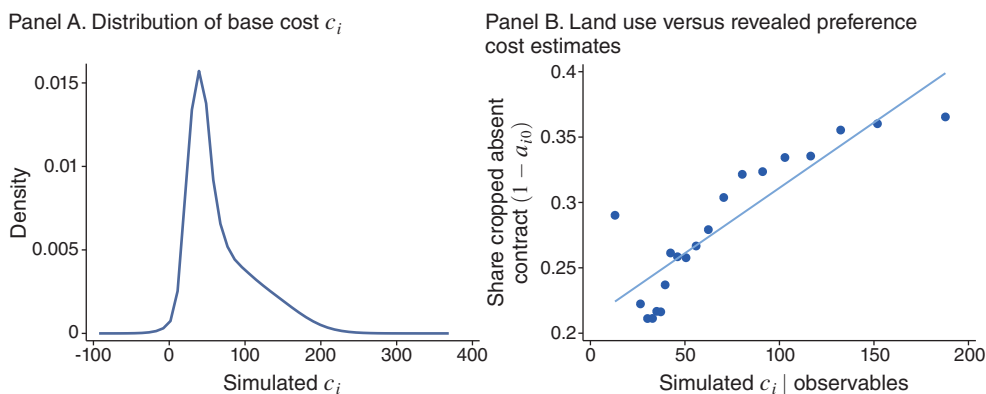


FIGURE 6. ESTIMATED LANDOWNER BASE COSTS

Notes: Panel A presents a kernel density plot of estimates of the base cost c_i of a CRP contract, pooling bidders across auctions. Panel B correlates base costs, c_i , conditional on observable characteristics $\mathbf{z}_i$, with land use outcomes measuring landowner additionality in the remote-sensing data. Panel B is restricted to the 2016 auction and the 82 percent of bidders who lose. $\mathbf{z}_i$ includes interactions of soil productivity, prior CRP, and prior land use. Costs are reported in dollars per acre per year.

landowners with higher base costs are more additional, indicating the presence of adverse selection mediated by observables.

Table 4 presents select parameter estimates in $\tau(\mathbf{z}_i, c_i, \kappa_i)$ that describe how additionality varies with (c_i, κ_i) . Each column of Table 4 presents a different specification of $\tau(\mathbf{z}_i, c_i, \kappa_i)$. Columns 1 and 2 restrict to a correlation between additionality and c_i and impose that $\alpha = 0$. Column 1 includes observable characteristics from the scoring rule, and column 2 adds observable characteristics that parameterize $F_{c,\kappa|\mathbf{z}}$ (interactions of soil productivity and prior land use). Consistent with the positive correlation between bids and additionality in Figure 5, panel A and Figure 5, panel B, the positive coefficients in columns 1 and 2 of Table 4 indicate that landowner additionality is systematically correlated with unobserved costs, conditional on observable characteristics. The coefficient estimates in Table 4 reflect the adverse selection presented in Figure 5, panel A and Figure 5, panel B.

Columns 3 and 4 allow additionality to also depend on κ_i . Column 3 only allows tree-related-action costs to impact additionality. The coefficient on tree-related κ_{ij} is positive and large, while the coefficient on c_i is reduced but still positive. Column 4 allows additionality to vary with κ_i more flexibly, and the residual correlation between costs and additionality loads onto κ_i . The largest coefficient remains on tree-related contracts.

We examine model fit in Supplemental Appendix E. For costs, we compare model-implied choices to data (Supplemental Appendix Figure E.6) and our estimates to some limited administrative data submitted to the USDA (Supplemental Appendix Table E.3). Our fit is reasonable, and model-implied costs are similar in rank and in magnitude to the administrative data. For $\tau(\mathbf{z}_i, c_i, \kappa_i)$, we examine the fit by correlating observations of $1 - a_{i0}$ and model-implied costs among rejected bidders (Supplemental Appendix Figure E.7); the linear functional form of $\tau(\mathbf{z}_i, c_i, \kappa_i)$ appears reasonable. Finally, we calculate a model-implied estimate of additionality at the RD margin of 23 percent, similar to our estimates in Section III.

TABLE 4—ADDITIONALITY AS A FUNCTION OF LANDOWNER COSTS

	Estimates of $\tau(\mathbf{z}_i, c_i, \kappa_i)$			
	(1)	(2)	(3)	(4)
β : coefficient on base cost (c_i)	0.0018 (0.0001)	0.0017 (0.0002)	0.0007 (0.0002)	−0.0008 (0.0003)
α : coefficient on top-up cost (κ_{ij})				
Trees			0.0025 (0.0003)	0.0047 (0.0004)
Native grasses				0.0020 (0.0004)
Habitat				0.0032 (0.0004)
Rare habitat				0.0034 (0.0005)
Wildlife food plot				0.0030 (0.0005)
Pollinator habitat				0.0022 (0.0005)
Includes $\mathbf{z}_i^s$	✓	✓	✓	✓
Includes prior land use and soil prod. $\mathbf{z}_i$		✓	✓	✓

Notes: Table presents select coefficient estimates of $\tau(\mathbf{z}_i, c_i, \kappa_i)$ (equation (14)). Coefficients measure how additionality varies with a \$1-per-acre-per-year change in costs. Positive coefficients indicate a positive correlation between costs and additionality. Parameter estimates obtained via MSM using the remote-sensing land use data (for losing bidders in the 2016 auction) and bids given simulated (c_i, κ_i) and optimal bidding in equation (10). All specifications include flexible controls for the components of the scoring rule excluding landowners’ WPZ and AQZ status. Columns 2–4 control for the 32 cells of soil productivity, prior CRP status, and prior cropping status that parameterize bidder costs. Standard errors calculated by block bootstrap (at the bidder level) and incorporate estimation error in costs by drawing from the estimated variance-covariance matrix of the parameters in $F_{c,\kappa|\mathbf{z}}$ in the bootstrap procedure.

D. Social Benefits

To calculate social welfare, we require estimates of the social benefits of contracted actions, $B_j(\mathbf{z}_i^s)$. We index $B_j(\mathbf{z}_i^s)$ by j to account for heterogeneous social benefits across contracts and allow $B_j(\mathbf{z}_i^s)$ to depend on observable characteristics in the scoring rule $\mathbf{z}_i^s$ that capture heterogeneity in the environmental sensitivity of landowners. We take average estimates of the value of CRP actions from literature that quantifies benefits from habitat restoration and reductions in erosion, water and air pollution, and greenhouse gas emissions (Johnson et al. 2016; Feather, Hellerstein, and Hansen 1999; Hansen 2007). We take relative valuations across characteristics $\mathbf{z}_i^s$ and contracts j from the scoring rule. This is justified by USDA reports that explicitly state that scores assume full additionality ($\tau = 1$) when scoring relative benefits (Claassen, Horowitz, Duquette, and Ueda 2014). Supplemental Appendix F provides details.

This approach makes important assumptions that add nuance to interpretation. First, the true social benefits of each action may diverge from the scoring rule, in which case our results could instead be interpreted as representing the revealed preferences of the USDA. Second, while we cannot rule out the possibility of some displaced conservation elsewhere, the CRP is targeted specifically at environmentally

sensitive land; our results could be interpreted as the net social benefits of targeting conservation in these environmentally sensitive locations.³⁵

Finally, we've specified additionality as one-dimensional (land retirement), but the menu of contracts is multidimensional. This is due to fundamental data limitations, the substantial simplification that focusing on only this one dimension affords, and the fact that the primary incentivized activity of the CRP is land retirement. Our baseline specification follows Section I and calculates contract value as $B_j(\mathbf{z}_i^s) \cdot \tau(\mathbf{z}_i, c_i, \kappa_i)$.³⁶ However, in Supplemental Appendix G, we present and discuss results under an alternative assumption where additionality only affects the base contract. This assumes a valuation of contracts equal to $B_0(\mathbf{z}_i^s) \cdot \tau(\mathbf{z}_i, c_i, \kappa_i) + B^j(\mathbf{z}_i^s)$, where $B_0(\mathbf{z}_i^s)$ is the value of the base action and $B^j(\mathbf{z}_i^s)$ is the incremental value of the top-up action.³⁷

V. Social Welfare and Alternative Market Designs

Estimates of the costs (c_i, κ_i) and expected social benefits $B_j(\mathbf{z}_i^s) \cdot \tau(\mathbf{z}_i, c_i, \kappa_i)$ of CRP contracts allow us to analyze social welfare and market design. Define the expected social surplus of contracting with landowner i for contract j as

$$(15) \quad B_j(\mathbf{z}_i^s) \cdot \tau(\mathbf{z}_i, c_i, \kappa_i) - c_i - \kappa_{ij}.$$

Equation (15) is based on $\tau(\mathbf{z}_i, c_i, \kappa_i)$, not $1 - a_{i0}$. Because current and counterfactual mechanisms screen only on $(\mathbf{z}_i, c_i, \kappa_i)$, using equation (15) for comparisons of social welfare is without loss.

A. Graphical Analysis

We first examine allocative efficiency and pricing in the context of our graphical framework with a uniform price and a single contract. In these analyses, we collapse heterogeneity to one dimension of cost for a single contract (as in Section I).

Figure 7 presents the empirical analogue of Figure 1, graphing the marginal cost (MC) and contract value ($B \cdot \tau$) curves for the base contract. We simulate the minimum cost to landowners of fulfilling the base contract to construct the MC curve. Then, we calculate the average $B_0(\mathbf{z}_i^s) \cdot \tau(\mathbf{z}_i, c_i, \kappa_i)$, where $B_0(\mathbf{z}_i^s)$ denotes the social benefit of the base action, at each quantile of this cost distribution to obtain the contract value curve. The two facts from Section III are reflected in Figure 7. The contract value curve lies below B , representing landowners who are not additional, and is upward sloping, representing adverse selection. Figure 7 illustrates the welfare implications of these two facts in the context of our stylized framework.

First, the contract value curve crosses the marginal cost curve from above, and social surplus—equal to the vertical distance between the contract value and marginal cost curves—is monotonically decreasing. The empirical market for the base

³⁵ Supplemental Appendix Figure G.1 illustrates alternative assumptions on $B_j(\mathbf{z}_i^s)$ in our graphical welfare framework.

³⁶ For tree actions, we can test this assumption because we can distinguish between trees and grasses in the remote-sensing data. We find that over 60 percent of rejected bidders choosing a tree contract have the majority of their bid land in trees, and 92 percent have some trees, supporting this as our baseline assumption.

³⁷ See Supplemental Appendix Figure G.7 for results under this alternative assumption.

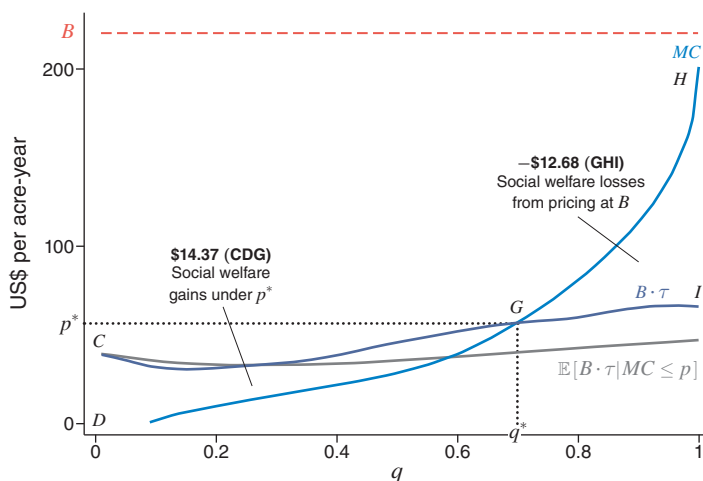


FIGURE 7. EMPIRICAL GRAPHICAL ANALYSIS

Notes: Figure presents the empirical version of Figure 1 for the base contract. The horizontal axis is the share of the population, ordered by costs of accepting a base CRP contract. Values are reported in dollars per acre per year. The MC curve is the inverse distribution function of the minimum cost to fulfill the base contract. B denotes the average value of the base contract action, calculated as described in Supplemental Appendix F. $B \cdot \tau$ denotes the incremental value of contracting, relative to no contract, averaged at each quantile of the population distribution of the base cost. The vertical distance between the $B \cdot \tau$ and MC curves represents the social surplus from contracting at each quantile of the population distribution of costs. The upward-sloping $B \cdot \tau$ curve illustrates the presence of adverse selection in the market. To illustrate adverse selection, figure also plots the average contract value curve (calculated as the average $B \cdot \tau$ of all landowners selecting into the market at any given price p , $\mathbb{E}[B \cdot \tau | MC \leq p]$). Both $B \cdot \tau$ and $\mathbb{E}[B \cdot \tau | MC \leq p]$ graphed with local polynomial regressions. The intersection of the MC and $B \cdot \tau$ curves denotes the socially optimal uniform price, p^* . Triangle CDG represents social welfare gains under the socially optimal price. The triangle GHI represents social welfare losses at B .

contract described by Figure 7 is therefore similar to Figure 1, panel A, not Figure 1, panel B. This socially optimal uniform price implements social welfare gains of \$14.37 per acre-year in region CDG.³⁸

Second, Figure 7 illustrates inefficient contracting when prices are set at B (the average $B_0(\mathbf{z}_i^s)$ across landowners), ignoring counterfactual land use. Pricing at B leads to social welfare losses of \$12.68 per acre-year in triangle GHI, almost all of the gains realized in triangle CDG.³⁹ These social welfare losses highlight the importance of quantitative analysis of the joint distribution of costs and additionality to set socially optimal incentives. This is underscored by the fact that the presence of adverse selection implies that additionality *at the margin* will be higher than the *average* additionality of the landowners in the market, plotted for illustration in Figure 7 in gray.

Heterogeneity.—Figure 8, panel A examines heterogeneity across the base and tree contracts motivated by the evidence of substantial adverse selection on tree contracts in Figure 5, panel C. Figure 8, panel A calculates the marginal cost curve

³⁸ Supplemental Appendix Figure G.2 demonstrates the robustness of the conclusions in Figure 7 and Figure 8, panel A to estimation error by plotting marginal cost and contract value curves across bootstrap draws.

³⁹ The landowners that make up triangle GHI incur costs from accepting the contract that are larger than the incremental social benefits from increased conservation (the $B \cdot \tau$ curve). These costs represent both the forgone option to crop, as the contract constrains them to conserve, and hassle costs associated with contracting (discussed in Section I). However, we do not distinguish between these two sources of costs.

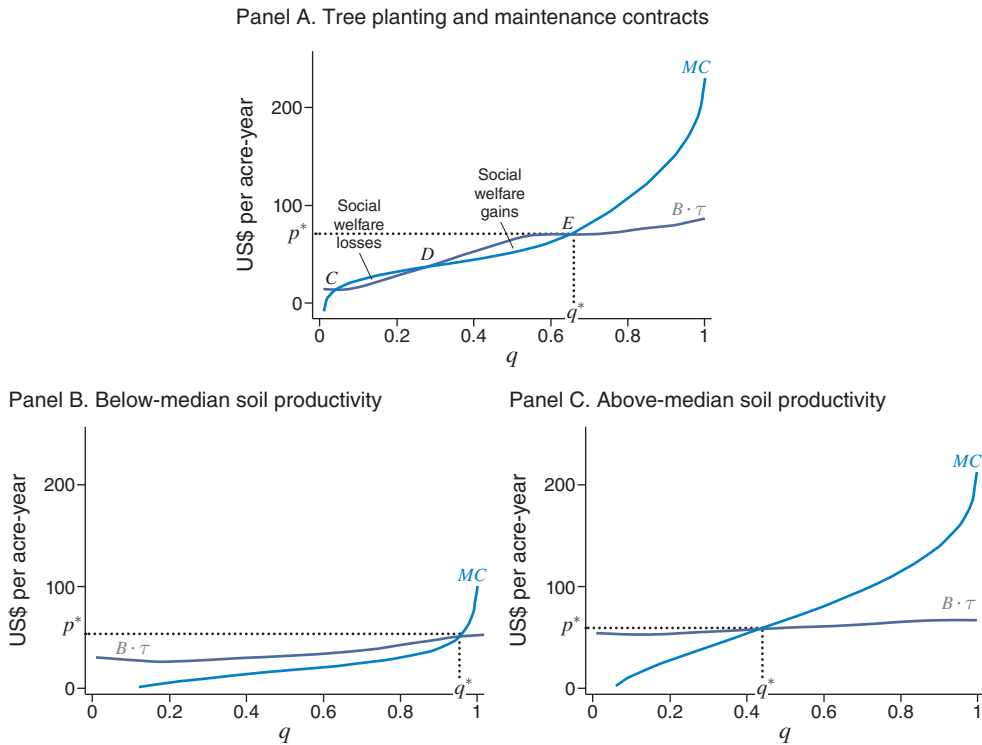


FIGURE 8. EMPIRICAL GRAPHICAL ANALYSIS: HETEROGENEITY ACROSS CONTRACTS AND OBSERVABLES

Notes: Figure presents the empirical version of Figure 1. In panel A, the MC curve is calculated as the inverse distribution function of the minimum cost to fulfill a tree planting and maintenance contract. The horizontal axis is the share of the population, ordered by costs of accepting a tree planting and maintenance contract. In panels B and C, the MC curve is calculated as the inverse distribution function of the minimum cost to fulfill the base contract, split by whether landowners are below (panel B) or above (panel C) median estimated soil productivity. The horizontal axis in panels B and C is the share of the population, ordered by costs of accepting the base contract, within these subpopulations. Values are reported in dollars per acre per year. $B \cdot \tau$ denotes the incremental value of contracting, relative to no contract, averaged at each quantile of the population distribution of cost of accepting a contract. The vertical distance between the $B \cdot \tau$ and MC curves represents the social surplus from contracting at each quantile of the population distribution of costs. The upward-sloping $B \cdot \tau$ curve illustrates the presence of adverse selection in the market. $B \cdot \tau$ graphed with local polynomial regressions. In panel A, the efficient allocation defined in equation (5) cannot be implemented. In panels B and C, the intersection of the MC and $B \cdot \tau$ curves denotes the socially optimal uniform price, p^* .

as the minimum cost required to fulfill any tree-related contract and plots the contract value at each quantile of this distribution. The exercise requires substantial extrapolation, but it illustrates how alternative landowner-type distributions across important classes of contracts in our setting can generate different conclusions.

In Figure 8, panel A, the contract value curve crosses the marginal cost curve from below, leading to social welfare losses at low q . The social welfare gains in region DE cannot be implemented without incurring losses among lower-cost landowners in CD. This is because, as in Figure 1, panel B, the ordering of landowners by social surplus (the vertical distance between the contract value and marginal cost curves) differs from the ordering of landowners by costs (the construction of the x -axis, q). Because they are less additional, the lowest-cost landowners are not the highest social value.

Motivated by Figure 5, panel D, Figure 8, panel B and Figure 8, panel C replicate Figure 7 in subpopulations split by estimated soil productivity. Both contract value and marginal cost curves differ across low- versus high-soil-productivity landowners, implying different socially optimal prices.

B. Alternative Auctions

The objective of PES mechanisms is to *impact conservation* at lowest cost. Mechanisms that consider reports of (c_i, κ_i) but not the effect of contracting on conservation $\tau(\mathbf{z}_i, c_i, \kappa_i)$ may not advance this goal. We simulate bidding and additionality under status quo and counterfactual auctions to investigate this possibility and the performance of alternative designs. Figure 9 and Table 5 present results. We calculate social welfare under each allocation:

$$(16) \quad \sum_i \sum_j (B_j(\mathbf{z}_i^s) \cdot \tau(\mathbf{z}_i, c_i, \kappa_i) - c_i - \kappa_{ij}) \cdot x_{ij}.$$

Table 5 also reports additional details: USDA spending, landowner surplus, environmental benefits $\sum_i \sum_j B_j(\mathbf{z}_i^s) \cdot \tau(\mathbf{z}_i, c_i, \kappa_i) \cdot x_{ij}$, average additionality, and the share of bidders allocated a contract.

Status Quo.—Because social welfare depends on additionality but the design of the status quo auction does not, the social value of the CRP is ambiguous. We document social welfare gains of \$120 million per auction in the status quo (bar (1) of Figure 9), across the seven auctions in our sample. This is calculated by simulating optimal bidding, given our estimates of costs and beliefs.

However, social welfare under the status quo auction is only 12 percent of an efficient benchmark. This efficient benchmark is defined as the allocation that uses all observables $\mathbf{z}_i$ and the full vector of costs (c_i, κ_i) to maximize social welfare (equation (16)) subject to two constraints. First, each landowner must obtain at most one contract $\sum_j x_{ij} \leq 1$. Second, no more landowners are allocated contracts, in each auction, than under the status quo. Because many landowners are not additional, the efficient allocation involves contracting with fewer landowners than the status quo, and the quantity constraint does not bind for most auctions.

This efficient allocation may not be implementable in an incentive-compatible auction if social surplus and allocations are not monotone in bidder costs (Myerson 1981). This complication is relevant because of adverse selection; once the mechanism's impact on conservation (additionality, or $\tau(\mathbf{z}_i, c_i, \kappa_i)$) is considered, the lowest-cost landowners may not be the highest social value. This issue is illustrated in principle in Figure 1, panel B and based on our estimates in Figure 8, panel A.

Alternative Auctions.—The status quo auction underperforms in part because it does not consider additionality in its design. Implementing the efficient allocation may be impossible, but changes to the status quo may reduce the gap. We consider an approach that treats landowners asymmetrically by the expected social benefits of a CRP contract, which depends on their additionality. This differs from alternative solutions that attempt to define eligibility requirements for participation.

Contracting with a low-expected-additionality landowner could be justified at sufficiently low cost, and landowners who are likely to be additional may still counterfactually conserve with some probability. We allow incentives to differ across landowner observables and contracts using counterfactual scoring rules based on predictions of $B_j(\mathbf{z}_i^s) \cdot \tau(\mathbf{z}_i, c_i, \kappa_i)$. These auctions build directly on the status quo, which uses a scoring rule based on $B_j(\mathbf{z}_i^s)$.

We define $s_j(\mathbf{z}_i)$ to score the expected social benefit of contracting with a bidder with characteristics $\mathbf{z}_i$ for contract j . We focus on linear rules based on (a slight simplification of) the functional form of the status quo scoring rule:

$$(17) \quad s_j(\mathbf{z}_i) = \omega_{\mathbf{z}} \cdot \mathbf{z}_i + \omega_j,$$

where $\omega_{\mathbf{z}}$ parameterizes scores across observables $\mathbf{z}_i$, and ω_j parameterizes scores across contracts j .

Our baseline counterfactuals implement allocations with a Vickrey–Clarke–Groves (VCG) mechanism in auctions that we term “Vickrey auctions with scoring.” These auctions maximize a definition of social welfare implied by the scoring rule

$$(18) \quad \sum_i \sum_j (s_j(\mathbf{z}_i) - c_i - \kappa_{ij}) \cdot x_{ij},$$

where bidders are treated asymmetrically by $s_j(\mathbf{z}_i)$, not $B_j(\mathbf{z}_i^s) \cdot \tau(\mathbf{z}_i, c_i, \kappa_i)$. In Vickrey auctions with scoring, bidders truthfully report their vector of (c_i, κ_i) , then are ranked by $\max_j s_j(\mathbf{z}_i) - c_i - \kappa_{ij}$.⁴⁰ The highest-scoring bidders subject to the auction’s quantity constraint are allocated the contract $\arg \max_j s_j(\mathbf{z}_i) - c_i - \kappa_{ij}$. Importantly, $s_j(\mathbf{z}_i)$ depends *only* on observable characteristics. Moreover, $s_j(\mathbf{z}_i)$ only includes immutable characteristics to avoid introducing perverse incentives to game the scoring rule. For these reasons, the scoring rule will *not* capture all heterogeneity in $\tau(\mathbf{z}_i, c_i, \kappa_i)$.

Vickrey auctions with scoring have both advantages and disadvantages. They isolate the design of the scoring rule by implementing an allocation that maximizes equation (18); this allocation is computationally cheap to compute, the market designer needs only to compute $s_j(\mathbf{z}_i)$, and bidders need only to report their costs. At the same time, the VCG mechanism is infrequently used and features important practical weaknesses (Ausubel and Milgrom 2005): It reveals a substantial amount of information, it can be expensive for the market designer, and it requires that bidders understand the incentives of an unfamiliar auction format.⁴¹ Therefore, while our baseline counterfactuals implement Vickrey auctions with scoring, we will also

⁴⁰ Supplemental Appendix G provides more details, including the incentive payments that implement this allocation.

⁴¹ However, because every bidder and every contract is a substitute, the Vickrey auctions with scoring do not exhibit many of the deficiencies discussed in Ausubel and Milgrom (2005). Moreover, the incentive payment for bidder i and contract j (if she wins the auction) is simple: $s_j(\mathbf{z}_i)$ net of the scoring rule implied social surplus of the marginal bidder.

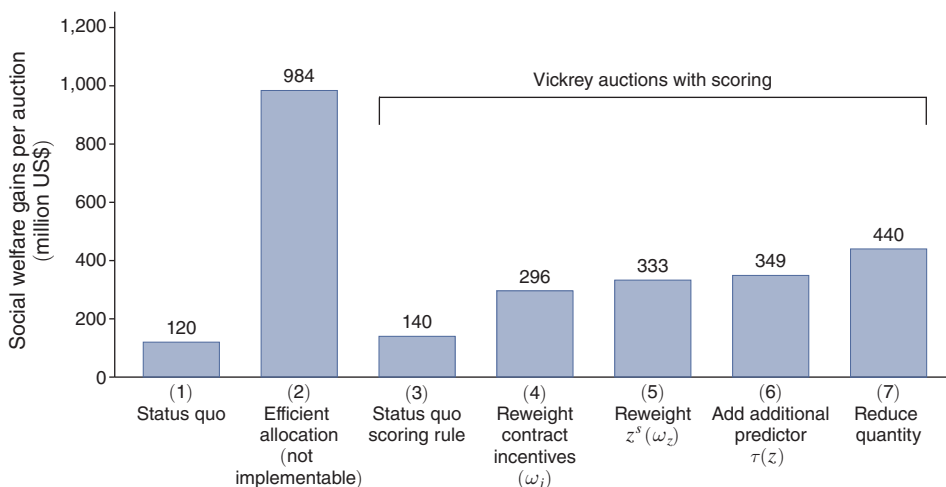


FIGURE 9. SOCIAL WELFARE UNDER ALTERNATIVE AUCTIONS

Notes: Figure presents estimates of the social welfare gains (defined in equation (16)) under status quo and alternative auctions. Results reported in million dollars per auction. All auctions impose that each landowner obtains at most one contract and that total contracts awarded cannot exceed the status quo, in each auction. Bar (1) simulates the status quo. Bar (2) calculates the social welfare gains under an efficient allocation that allocates contracts using all $\mathbf{z}_i$ and (c_i, κ_i) to maximize equation (16). Due to adverse selection, this allocation may not be implementable. Bars (3)–(7) calculate social welfare under alternative Vickrey auctions with scoring (see Section V for more details). Bars (3)–(6) hold quantity (the number of landowners allocated contracts) constant at the status quo and change the scoring rule $s_j(\mathbf{z}_i)$ defined in equation (17). Bar (3) uses the existing scoring rule $s_j(\mathbf{z}_i) = B_j(\mathbf{z}_i^s)$. Bar (4) uses a scoring rule with the social-surplus-maximizing incentives across contracts (ω_j). Bar (5) uses a scoring rule with the social-surplus-maximizing asymmetry terms across bidders using characteristics already in the scoring rule ($\mathbf{z}_i^s$). Bar (6) adds an additional characteristic to the scoring rule, a prediction of $\tau(\mathbf{z}_i, c_i, \kappa_i)$ based on immutable characteristics of landowners already collected by the USDA (deciles of soil productivity and wind and water erosion). Bar (7) uses the same scoring rule as bar (6) but reduces the number of contracts allocated to landowners: Only landowners with positive scoring-rule-implied social surplus $\max_j s_j(\mathbf{z}_i) - c_i - \kappa_{ij} \geq 0$ are awarded contracts.

assess the performance of alternative scoring rules in mechanisms currently in use by the USDA.

Results: Vickrey Auction with Scoring.—Figure 9 plots the social welfare gains (evaluated using equation (16)) under progressively improving Vickrey auctions with scoring. The auctions in bars (3)–(6) vary $s_j(\mathbf{z}_i)$, holding the number of awarded contracts fixed at the status quo quantity for each auction. Additional details are reported in the corresponding columns of Table 5.

Bar (3) maintains the status quo scoring rule, $s_j(\mathbf{z}_i) = B_j(\mathbf{z}_i^s)$, but changes the auction mechanism to VCG. This counterfactual isolates the impact of a scoring rule that is naive to additionality (bar (2) versus bar (3)) and provides a basis for further comparisons that change only the rule $s_j(\mathbf{z}_i)$ but hold constant the VCG auction mechanism. If all landowners were additional, the scoring rule defined by $s_j(\mathbf{z}_i) = B_j(\mathbf{z}_i^s)$ would implement the efficient allocation. Instead, it implements only 14 percent of social welfare gains under the efficient allocation.

Bars (4)–(6) in Figure 9 adjust the scoring rule $s_j(\mathbf{z}_i)$ defined in equation (17) based on predictions of additionality. First, in bar (4), we adjust the scoring of the menu of contracts, ω_j , to maximize equation (16), holding ω_z constant at the status

TABLE 5—OUTCOMES UNDER ALTERNATIVE AUCTIONS

	Vickrey auctions with scoring								Status quo auction	Vickrey auction with scoring 2016 only (hold out)
	Status quo	Efficient allocation	Status quo rule	Reweight contracts (ω_j)	Reweight $\mathbf{z}_i^s$ ($\omega_{\mathbf{z}}$)	Add $\hat{\tau}(\mathbf{z}_i)$	Reduce quantity	Uniform multiplier	Best rule	Limited-info rule
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>Panel A. Welfare and spending (million \$ per auction)</i>										
Social welfare	120	984	140	296	333	349	440	325	236	217
USDA spending	1,615		2,419	2,146	2,250	2,033	877	2,225	1,373	100
Landowner surplus	666		1,470	1,430	1,559	1,329	523	1,543	596	61
Environmental value	1,069	1,461	1,089	1,012	1,024	1,052	794	1,007	1,013	256
<i>Panel B. Other outcomes</i>										
Additionality contract	0.201	0.447	0.197	0.198	0.203	0.209	0.208	0.201	0.205	0.305
Share awarded contract	0.81	0.50	0.81	0.81	0.81	0.81	0.62	0.81	0.81	0.18

Notes: Table presents results under current and alternative auctions. Panel A tabulates social welfare (equation (16)), USDA spending, landowner surplus, and environmental value $\sum_i \sum_j B_j(\mathbf{z}_i^s) \cdot \tau(\mathbf{z}_i, c_i, \kappa_i) \cdot x_{ij}$ in million dollars per auction. Panel B tabulates average additionality of contracting landowners and the share of landowners with a contract. All auctions impose that each landowner obtains at most one contract and that total contracts awarded cannot exceed the status quo, in each auction. Columns 1–9 present results for all auctions in the sample. Column 10 presents results for the 2016 auction only. Column 1 simulates the status quo. Column 2 simulates an efficient allocation that allocates contracts using all $\mathbf{z}_i$ and (c_i, κ_i) to maximize equation (16). Due to adverse selection, this allocation may not be implementable. Columns 3–8 simulate alternative Vickrey auctions with scoring (see Section V for more details). Columns 3–6 and 8 and 9 hold quantity (the number of landowners allocated contracts, in each auction) constant at the status quo and change the scoring rule $s_j(\mathbf{z}_i)$ defined in equation (17). Column 3 uses the existing scoring rule $s_j(\mathbf{z}_i) = B_j(\mathbf{z}_i^s)$. Column 4 uses a scoring rule with the social-surplus-maximizing incentives across contracts (ω_j). Column 5 uses a scoring rule with the social-surplus-maximizing asymmetry terms across bidders using characteristics already in the scoring rule ($\mathbf{z}_i^s$). Column 6 adds an additional characteristic to the scoring rule, a prediction of $\tau(\mathbf{z}_i, c_i, \kappa_i)$ using immutable characteristics of landowners already collected by the USDA but not all included in the status quo scoring rule (deciles of soil productivity and wind and water erosion). Column 7 uses the same scoring rule as column 6 but reduces the number of contracts allocated to landowners: Only landowners with positive scoring-rule-implied social surplus $\max_j s_j(\mathbf{z}_i) - c_i - \kappa_{ij} \geq 0$ are awarded contracts. Column 8 uses a scoring rule $s_j(\mathbf{z}_i^s) = \theta \cdot B_j(\mathbf{z}_i^s)$ for a uniform multiplier θ that maximizes equation (16). Column 9 implements the scoring rule in columns 6 and 7 in the status quo auction format. Column 10 implements a Vickrey auction with scoring using the limited-information scoring rule. This rule is constructed exploiting cross-tract and cross-auction heterogeneity for pre-2016 auctions, measured in the administrative data (versus the remotely sensed data used in estimation). Evaluation in column 10 uses the holdout 2016 auction only (used to estimate the parameters in $\tau(\mathbf{z}_i, c_i, \kappa_i)$).

quo rule.⁴² This change to the scoring rule more than doubles the social welfare gains of the auction (an increase of \$156 million per auction). Next, in bars (5) and (6), we adjust the bidder asymmetry terms across observables, $\omega_{\mathbf{z}}$. Bar (5) first reweights the existing characteristics in the scoring rule, $\mathbf{z}_i^s$, based also on additionality. The final adjustment (bar (6)) adds an additional characteristic to the rule: an “additionality factor” $\hat{\tau}(\mathbf{z}_i)$, which we calculate by projecting $\tau(\mathbf{z}_i, c_i, \kappa_i)$ on immutable characteristics of landowners already collected by the USDA but not all incorporated in the status quo scoring rule (deciles of soil productivity and wind and water erosion). Then, we calculate the social-welfare-maximizing score using $\mathbf{z}_i^s$ and $\hat{\tau}(\mathbf{z}_i)$ added linearly as asymmetry terms in equation (17). This change only considers immutable characteristics. Other observables—for example, prior land use—would yield further gains but introduce incentives to game the rule. Together, these changes to bidder asymmetry terms yield a further \$53 million of social welfare gains per auction.

⁴²We solve for the ω_j that maximize equation (16) given simulations of landowner (c_i, κ_i) , estimates of $\tau(\mathbf{z}_i, c_i, \kappa_i)$, calibrations of $B_j(\mathbf{z}_i^s)$, and the allocation rule, holding $\omega_{\mathbf{z}}$ fixed at the status quo scoring rule.

Beyond the allocation rule, additionality also impacts the socially optimal size of the market. Because landowners are not all additional, the status quo quantity procured is *higher* than is socially optimal. In bar (7), we use our alternative scoring rules to reduce the size of the market, keeping the scoring rule $s_j(\mathbf{z}_i)$ of bar (6) but rejecting landowners with $\max_j s_j(\mathbf{z}_i) - c_i - \kappa_{ij} < 0$. This reduction in market size increases social welfare substantially, by \$91 million per auction.⁴³

The alternative auctions in Figure 9 outperform the status quo by adjusting the scoring rule to reflect the social benefit of contracting. This occurs via two channels. First, the status quo rule $B_j(\mathbf{z}_i^s)$ overweights asymmetry across characteristics $\mathbf{z}_i^s$ and contracts j . The heterogeneous social benefits of conservation $B_j(\mathbf{z}_i^s)$ are not fully realized when conservation would have counterfactually occurred. Second, bidders may be systematically heterogeneous in their additionality, which can be exploited in the scoring rule to align the allocation implemented by the auction with the socially optimal allocation that considers heterogeneity in both additionality and costs.

Column 8 in Table 5 explores these two mechanisms further. Holding quantity constant, we report outcomes under a scoring rule $s_j(\mathbf{z}_i) = \theta \cdot B_j(\mathbf{z}_i^s)$ for the (single) multiplier θ chosen to maximize equation (16). This examines the social welfare gains achieved with only a uniform instrument to adjust the scoring rule for additionality, in comparison to auctions that adjust ω_j and ω_z to reflect heterogeneously additional landowners (column 6). Adjusting the scoring rule based on heterogeneity in additionality yields social welfare gains of \$24 million per auction (column 6 versus column 8), but a uniform adjustment for additionality achieves a large share of the gains relative to the status quo. This highlights the importance of incorporating additionality to obtain the correct marginal rate of substitution between the social benefits and costs of each CRP contract.

Implementation in Practice.—The budgetary implications of alternative auctions are also relevant in practice. Holding quantity fixed, Vickrey auctions with scoring require greater USDA spending than the status quo (see column 3 versus column 1 of Table 5). However, the auction with the greatest social welfare gains also *reduces* USDA spending relative to the status quo (column 7 of Table 5) because reducing the size of the market, and therefore total USDA spending, increases social welfare. Still, across auctions, government spending exceeds the value of environmental services procured, $\sum_i \sum_j B_j(\mathbf{z}_i^s) \cdot \tau(\mathbf{z}_i, c_i, \kappa_i) \cdot x_{ij}$, due to the presence of adverse selection: The marginal landowner has a higher value of $\tau(\mathbf{z}_i, c_i, \kappa_i)$ than the inframarginal landowner.

The USDA may be hesitant to adopt a new auction format, particularly if it increases spending at the current program size. In column 9 of Table 5, we therefore implement our preferred scoring rule (column 6) in the status quo auction mechanism. We hold constant bidder beliefs about quantity procured and the number of bidders and iteratively update beliefs $G(S)$ (using our resampling procedure) and

⁴³ Further differences between bar (7) and bar (2) reflect (i) $\mathbf{z}_i$ that are not incorporated into the scoring rule to avoid perverse incentives to game the rule, (ii) private landowner costs in $\tau(\mathbf{z}_i, c_i, \kappa_i)$, and (iii) the functional form of equation (17) relative to $B_j(\mathbf{z}_i^s) \cdot \tau(\mathbf{z}_i, c_i, \kappa_i)$.

bidders' best responses (given $G(S)$).⁴⁴ Adjusting the scoring rule doubles social welfare gains relative to the status quo and does not increase USDA spending. This provides another useful comparison: Holding fixed the scoring rule, the switch to VCG yields gains of only \$20 million per auction (column 3 versus column 1 of Table 5), while holding fixed the auction format, the alternative scoring rule yields gains over five times as large (column 9 versus column 1).

Our approach also assumes that the function $\tau(\mathbf{z}_i, c_i, \kappa_i)$ is stable and that the distribution of costs is known. If either of these assumptions are violated, potential efficiency gains may be limited. To address this concern, we construct an alternative limited-information scoring rule that can be computed using only observables and can be evaluated in a holdout auction to assess stability (see Supplemental Appendix G for details).⁴⁵ In contrast to our baseline approach, this alternative scoring rule estimates heterogeneity in additionality by observables, matching cross-tract heterogeneity in cropping rates pre-auction (Supplemental Appendix Figure D.7) and RD coefficients across auctions (Supplemental Appendix Table D.2). This allows us to use data on auctions pre-2016 to construct the scoring rule and evaluate its performance in the 2016 auction.⁴⁶ We use the administrative data in this exercise, instead of the remote-sensing data, to further assess the stability of $\tau(\mathbf{z}_i, c_i, \kappa_i)$ to the dataset used for estimation. Column 10 of Table 5 presents results, implementing the limited-information scoring rule with a Vickrey auction with scoring, evaluated in 2016 using our estimates of $F_{c,\kappa|\mathbf{z}}$ and $\tau(\mathbf{z}_i, c_i, \kappa_i)$. We estimate substantial social welfare gains, even under this limited information rule: \$217 million, compared to status quo social welfare gains of \$120 million on average and \$91 million in 2016 (see Supplemental Appendix Figure G.6). We also obtain accurate estimates of the socially optimal market size under the limited information rule.⁴⁷

Finally, we assume the objective in equation (16), but the USDA may alternatively place higher weight on landowner surplus versus environmental objectives. We investigate this possibility by comparing the relative performance of our highest-performing auction (bar (7)) versus the status quo, varying the relative weights on landowner surplus versus environmental services procured (net of expenditures). The status quo becomes rationalized as preferred to our highest-performing auction when each dollar to landowners is worth more than three times a dollar spent toward conservation (see Supplemental Appendix Figure G.4 for more detail).⁴⁸

⁴⁴ Supplemental Appendix G provides details and also implements a series of budget-conscious posted-prices mechanisms, as used in other USDA programs including continuous CRP (results reported in Supplemental Appendix Table G.1).

⁴⁵ A motivation for this approach is that it could be feasibly implemented by the USDA for future auctions.

⁴⁶ The 2016 auction serves as our holdout auction because estimation of $\tau(\mathbf{z}_i, c_i, \kappa_i)$ uses observations of rejected bidders from the 2016 auction (with the most rejected bidders).

⁴⁷ See Supplemental Appendix Figure G.6, which also includes further comparisons between the performance of the limited-information and baseline rules in both the 2016 auction and in all auctions. Supplemental Appendix Figure G.5 compares the limited-information- and baseline-scoring rules.

⁴⁸ Alternatively, transfers may be costly, even given a quantity constraint. Supplemental Appendix Figure G.3 evaluates social welfare with a cost of funds, motivated by the need to finance expenditures with distortionary taxation.

VI. Conclusion

Questions related to additionality pose a central challenge to many domains of environmental market design. This paper combines data and theory to document this challenge, quantify its implications for social welfare, and evaluate alternative market designs in the largest auction mechanism for ecosystem services in the world. Linking satellite data to auction bids, we use an RD design to demonstrate that only one-quarter of marginal auction winners are additional. The lowest bidders are also more likely to counterfactually conserve, introducing adverse selection in the market. To quantify the implications of these facts and test possible remedies, we develop and estimate a joint model of bidding and land use that incorporates adverse selection on additionality.

A common approach to additionality is to define eligibility requirements that restrict who and what can trade; we propose a more flexible alternative. Because many markets will inevitably attract landowners who are with some probability not additional, allocation mechanisms should consider this dimension of heterogeneity in their design. We show how auctions can use scoring rules to cost-effectively impact conservation, selecting participants based on both expected additionality and costs, despite the fact that not all landowners in the market are additional. While our analysis focused exclusively on the CRP and the design of its auction, additionality also plays a central role in the performance and design of offset markets. In these markets, demand and intermediaries interact with supply-side adverse selection, presenting an interesting and impactful avenue for future research.

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